

Quantum Field Theory in Curved Spacetime

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These notes are not endorsed by the lecturers. All errors are mine.

Contents

1. Introduction	4
2. Note on General Relativity	5
2.1. The Einstein-Hilbert action	5
3. Canonical Quantization	6
3.1. Classical field theory	6
3.2. Quantization	7
4. The Casimir Effect	9
4.1. The effect	9
4.2. The energy	10
4.3. A note on the cosmological constant	12
5. Fields in Curved Space	14
5.1. The scalar field	14
5.2. Bogolubov transformations	16
5.3. Killing vectors	17
6. The Friedmann-Robertson-Walker Spacetime	20
6.1. The scalar field	20
6.2. Quantization	22
6.3. The instantaneous vacuum	23
6.4. The adiabatic vacuum	24
7. The FRW cosmology	25
7.1. The Big Bang	27
7.2. The horizon problem	28
8. Inflation	30
8.1. Slow-roll inflation	32
9. Field Perturbations	35
9.1. Quantum fluctuations of a scalar field in quasi de Sitter	35
9.2. The $m = 0$ scalar field in pure de Sitter	37
9.3. The scalar field in quasi de Sitter	39
9.4. The power spectrum	39
10. Metric Perturbations ¹	42
10.1. The linear metric	42
10.2. Gauge transformations	43
10.3. Gauge invariants	44
10.4. The longitudinal gauge	46

¹See <https://arxiv.org/pdf/hep-ph/0210162> for a review.

11. The Unruh Effect	50
11.1. Rindler spacetime	50
11.2. Quantum fields in Rindler space	51
11.3. The Unruh temperature	54
12. Hawking Radiation	57
12.1. Rindler-esque derivation of Hawking radiation	57
12.2. in de Sitter spacetime	58
12.3. Thermodynamics of black holes	59
12.4. Consequences	60

1. Introduction

We want to understand quantum field theory in the presence of a non-trivial gravitational field. The simplest approach to do this is by treating the gravitational field as a classical background on which quantum fields live. The majority of these notes cover this approach. Toward the end we do consider quantum fluctuations on the background itself.

The effective field theory approach to quantum gravity we use turns out sufficient to describe e.g. the early Universe and quantum effects near black holes. A fuller description would require a more sophisticated theory like string theory.

2. Note on General Relativity

The object of interest in general relativity is the *metric* field $g_{\mu\nu}$. To determine it we solve the *Einstein field equations*

$$\underbrace{\mathcal{R}_{\mu\nu} - \frac{1}{2}g_{\mu\nu}\mathcal{R}}_{\text{geometry of spacetime}} = 8\pi G_N \underbrace{T_{\mu\nu}}_{\text{matter}}$$

where $T_{\mu\nu}$ is the *energy-momentum tensor* and $\mathcal{R}_{\mu\nu}$ is the *Ricci tensor* with $\mathcal{R} \equiv g^{\mu\nu}\mathcal{R}_{\mu\nu}$ being the *Ricci scalar*. The Ricci tensor is defined by

$$\mathcal{R}_{\mu\kappa} \equiv g^{\lambda\nu}\mathcal{R}_{\lambda\mu\nu\kappa}$$

with

$$\mathcal{R}^{\sigma}{}_{\mu\nu\kappa} \equiv \partial_{\kappa}\Gamma^{\sigma}{}_{\mu\nu} - \partial_{\nu}\Gamma^{\sigma}{}_{\mu\kappa} + \Gamma^{\lambda}{}_{\mu\nu}\Gamma^{\sigma}{}_{\kappa\lambda} - \Gamma^{\lambda}{}_{\mu\kappa}\Gamma^{\sigma}{}_{\nu\lambda}$$

being the *Riemann-Christoffel curvature tensor*. Where

$$\Gamma^{\lambda}{}_{\mu\nu} \equiv \frac{1}{2}g^{\lambda\sigma}(\partial_{\mu}g_{\sigma\nu} + \partial_{\nu}g_{\sigma\mu} - \partial_{\sigma}g_{\mu\nu})$$

is the *Christoffel symbol* or *affine connection*.

2.1. The Einstein-Hilbert action

We would like some action

$$S_H = \int d^4x \sqrt{-g} \mathcal{L}_H$$

with $\mathcal{L}_H$ being a scalar and $g \equiv \det g_{\mu\nu}$. The simplest choice is $\mathcal{L}_H = \mathcal{R}$. We minimize this with respect to $g_{\mu\nu}$ by $\delta S_H = 0$ giving

$$\frac{1}{\sqrt{-g}} \frac{\delta S_H}{\delta g^{\mu\nu}} = \mathcal{R}_{\mu\nu} - \frac{1}{2}\mathcal{R}g_{\mu\nu} = 0$$

To include matter we add S_M

$$S = \frac{1}{8\pi G_N} S_H + S_M$$

We obtain

$$\frac{1}{\sqrt{-g}} \frac{\delta S}{\delta g^{\mu\nu}} = \frac{1}{8\pi G_N} \left[\mathcal{R}_{\mu\nu} - \frac{1}{2} g_{\mu\nu} \mathcal{R} \right] + \frac{1}{\sqrt{-g}} \frac{\delta S_M}{\delta g^{\mu\nu}} \stackrel{!}{=} 0$$

Leading us to the definition of the energy-momentum tensor

$$T_{\mu\nu} = -\frac{1}{\sqrt{-g}} \frac{\delta S_M}{\delta g^{\mu\nu}}$$

As a final remark we can always add another constant to $\mathcal{L}_H$ by $\mathcal{R} \rightarrow \mathcal{R} - 2\Lambda$ giving an additional term

$$\mathcal{R}_{\mu\nu} - \frac{1}{2} g_{\mu\nu} \mathcal{R} + \Lambda g_{\mu\nu} = 0$$

We call Λ the cosmological constant. We could also treat this as the energy density of vacuum by

$$\mathcal{R}_{\mu\nu} - \frac{1}{2} g_{\mu\nu} \mathcal{R} = \underbrace{-\Lambda g_{\mu\nu}}_{T_{\mu\nu}}$$

3. Canonical Quantization

3.1. Classical field theory

Before attempting to understand quantum field theory on a curved background we should understand quantum field theory on a flat background.

We consider the theory of a real scalar field $\phi(\mathbf{x}, t)$. This is described by

$$\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2$$

Which gives the *Klein-Gordon equation*

$$(\partial_\mu \partial^\mu + m^2) \phi(\mathbf{x}, t) = 0$$

To quantize our field we need the Hamiltonian formulation. We define the *canonical conjugate field* π by

$$\pi(\mathbf{x}, t) \equiv \frac{\partial \mathcal{L}}{\partial \dot{\phi}(\mathbf{x}, t)} = \dot{\phi}(\mathbf{x}, t)$$

Then we can write

$$\begin{aligned}
H &= \int d^3x [\pi(\mathbf{x}, t) \dot{\phi}(\mathbf{x}, t) - \mathcal{L}] \\
&= \frac{1}{2} \int d^3x [\pi^2 + (\nabla \phi)^2 + m^2 \phi^2]
\end{aligned}$$

Which is just the Legendre transform.

3.2. Quantization

We promote ϕ and π to operators, and impose the *continuous* generalization of the usual commutation relations

$$\begin{aligned}
[\phi(\mathbf{x}, t), \pi(\mathbf{y}, t)] &= i\delta^{(3)}(\mathbf{x} - \mathbf{y}) \\
[\phi(\mathbf{x}, t), \phi(\mathbf{y}, t)] &= [\pi(\mathbf{x}, t), \pi(\mathbf{y}, t)] = 0
\end{aligned}$$

We would like to determine the spectrum of our Hamiltonian. How we proceed is in no way obvious. We try expanding ϕ as

$$\phi(\mathbf{x}, t) = \int \frac{d^3\mathbf{k}}{(2\pi)^{3/2}} e^{i\mathbf{k}\cdot\mathbf{x}} \phi_{\mathbf{k}}(t)$$

with $\phi_{\mathbf{k}}^* = \phi_{-\mathbf{k}}$. Then the Klein-Gordon equation becomes

$$\left[\frac{\partial^2}{\partial t^2} + (\mathbf{k}^2 + m^2) \right] \phi_{\mathbf{k}}(t) = 0$$

implying each mode becomes a harmonic oscillator with frequency $\omega_{\mathbf{k}} \equiv \sqrt{\mathbf{k}^2 + m^2}$! We can now write our Hamiltonian as

$$H = \frac{1}{2} \int d^3k [\pi_{\mathbf{k}} \pi_{-\mathbf{k}} + \omega_{\mathbf{k}}^2 \phi_{\mathbf{k}} \phi_{-\mathbf{k}}]$$

Since every mode is a harmonic oscillator it is convenient to introduce ladder operators

$$\begin{aligned}
a_{\mathbf{k}}(t) &\equiv \sqrt{\frac{\omega_{\mathbf{k}}}{2}} \left(\phi_{\mathbf{k}} + i \frac{\pi_{\mathbf{k}}}{\omega_{\mathbf{k}}} \right) \\
a_{\mathbf{k}}^\dagger(t) &\equiv \sqrt{\frac{\omega_{\mathbf{k}}}{2}} \left(\phi_{-\mathbf{k}} - i \frac{\pi_{-\mathbf{k}}}{\omega_{\mathbf{k}}} \right)
\end{aligned}$$

The equations of motion for ϕ and π become (since $\dot{\phi}_{\mathbf{k}} = \pi_{\mathbf{k}}$)

$$\frac{d}{dt}a_{\mathbf{k}}^{\dagger} = i\omega_{\mathbf{k}}a_{\mathbf{k}}^{\dagger}; \quad \frac{d}{dt}a_{\mathbf{k}} = -i\omega_{\mathbf{k}}a_{\mathbf{k}}$$

Which has the solution

$$a_{\mathbf{k}}^{\dagger}(t) = a_{\mathbf{k}}^{\dagger}(0)e^{i\omega_{\mathbf{k}}t}; \quad a_{\mathbf{k}}(t) = a_{\mathbf{k}}(0)e^{-i\omega_{\mathbf{k}}t}$$

Where the ladder operators must satisfy

$$[a_{\mathbf{k}}, a_{\mathbf{k}'}^{\dagger}] = \delta^{(3)}(\mathbf{k} - \mathbf{k}')$$

$$[a_{\mathbf{k}}^{\dagger}, a_{\mathbf{k}'}^{\dagger}] = [a_{\mathbf{k}}, a_{\mathbf{k}'}] = 0$$

We can now write

$$\phi_{\mathbf{k}} = \frac{1}{\sqrt{2\omega_{\mathbf{k}}}}(a_{\mathbf{k}}e^{-i\omega_{\mathbf{k}}t} + a_{-\mathbf{k}}^{\dagger}e^{i\omega_{\mathbf{k}}t})$$

$$\pi_{\mathbf{k}} = i\sqrt{\frac{\omega_{\mathbf{k}}}{2}}(a_{-\mathbf{k}}^{\dagger}e^{i\omega_{\mathbf{k}}t} - a_{\mathbf{k}}e^{-i\omega_{\mathbf{k}}t})$$

Then our Hamiltonian becomes

$$\begin{aligned} H &= \int d^3k \frac{\omega_{\mathbf{k}}}{2} [a_{\mathbf{k}}a_{\mathbf{k}}^{\dagger} + a_{\mathbf{k}}^{\dagger}a_{\mathbf{k}}] \\ &= \int d^3k \left[\omega_{\mathbf{k}}a_{\mathbf{k}}^{\dagger}a_{\mathbf{k}} + \underbrace{\frac{\omega_{\mathbf{k}}}{2}\delta^{(3)}(0)}_{\substack{\text{vacuum energy} \\ \rightarrow \infty}} \right] \end{aligned}$$

We can easily compute

$$[H, a_{\mathbf{k}}^{\dagger}] = \omega_{\mathbf{k}}a_{\mathbf{k}}^{\dagger}; \quad [H, a_{\mathbf{k}}] = -\omega_{\mathbf{k}}a_{\mathbf{k}}$$

Vacuum is defined by $a_{\mathbf{k}}|0\rangle = 0$ for all $\mathbf{k}$. Any excited state can then be built by acting with $a^{\dagger}$ on $|0\rangle$. To see this consider

$$H(a_{\mathbf{k}}^{\dagger}|0\rangle) = \omega_{\mathbf{k}}a_{\mathbf{k}}^{\dagger}|0\rangle + \underbrace{a_{\mathbf{k}}^{\dagger}H|0\rangle}_{\text{dropping infinity}}$$

This is consistent with interpreting $|\mathbf{k}\rangle = a_{\mathbf{k}}^{\dagger}|0\rangle$ as a one-particle state with momentum $\mathbf{k}$ and energy $E = \omega_{\mathbf{k}}$.

We can now express $\phi(\mathbf{x}, t)$ as

$$\phi(\mathbf{x}, t) = \int \frac{d^3k}{(2\pi)^{3/2}} \frac{1}{\sqrt{2\omega_k}} \left[a_{\mathbf{k}} e^{-i\omega_k t + i\mathbf{k} \cdot \mathbf{x}} + a_{\mathbf{k}}^\dagger e^{i\omega_k t - i\mathbf{k} \cdot \mathbf{x}} \right]$$

Typically this form is written immediately (with commutation relations for a and $a^\dagger$) to *shortcut* quantization.

4. The Casimir Effect

Assuming vacuum energy acts like any other energy then according to relativity it will gravitate. The vacuum energy density is infinite so this implies an infinite gravitational field! However since we exist something about this is wrong.

Above we found

$$\begin{aligned} E_0 &= \langle 0 | H | 0 \rangle \\ &= \int d^3k \frac{\omega_k}{2} \delta^{(3)}(0) \sim \text{diverges} \end{aligned}$$

which is problematic. We might argue the integral has an upper limit since particles with λ_{Compton} smaller than the Planck length l_p would become black holes since their Schwarzschild radius r_s would be larger than λ_{Compton} . However introducing a UV cutoff $\Lambda \sim M_p$ still yields $\varepsilon_0 \sim M_p^4$ which is enormous. This is called *regularising* the integral and we see the divergence reappears if we remove the cutoff by $\Lambda \rightarrow \infty$. Typically regularisation is followed by *renormalisation* which is simply a prescription for how to deal with the large arbitrary cutoff dependent contribution to some physical quantity. Above this quantity is the cosmological constant.

4.1. The effect

The *Casimir effect* is an experimentally verified prediction of the above procedure which we now derive. Consider a massless scalar field ϕ in Minkowski $1 + 1$ space-time. We now consider two plates separated by a distance L . We require the field vanishes on the plates

$$\phi|_{x=0} = \phi|_{x=L} = 0$$

and consider solutions to the equation of motion

$$\partial_t^2 \phi - \partial_x^2 \phi = 0$$

These are given by

$$\phi = \sum_{n=1}^{\infty} (A_n e^{-i\omega_n t} + B_n e^{i\omega_n t}) \sin \omega_n x$$

where $\omega_n \equiv n\pi/L$.

Without the plates we would have the following mode expansion

$$\phi = \int \frac{dk}{\sqrt{2\pi}} \frac{1}{\sqrt{2\omega_k}} [a_k e^{-i\omega_k t + ikx} + a_k^\dagger e^{i\omega_k t - ikx}]$$

However the plane wave modes are no longer viable after introducing the plates. The plates lead to a discrete subset of orthonormal functions being the basis for the mode expansion. We introduce the basis

$$g_n(x) = \sqrt{\frac{2}{L}} \sin \omega_n x; \quad \int_0^L g_m(x) g_n(x) dx = \delta_{mn}$$

The continuous k has been replaced by the discrete ω_n and now the basis satisfies the boundary conditions. We can now expand the field in terms of $e^{-i\omega_n t} g_n(x)$ instead of plane waves to obtain

$$\phi = \sqrt{\frac{2}{L}} \sum_{n=1}^{\infty} \frac{\sin \omega_n x}{\sqrt{2\omega_n}} [a_n e^{-i\omega_n t} + a_n^\dagger e^{i\omega_n t}]$$

We see that fewer modes are allowed between the plates as compared to outside. The only modes allowed are those that form standing waves. This leads to the vacuum energy between the plates being smaller as compared to outside!

4.2. The energy

We want to compute the vacuum energy between the plates. Using

$$H = \frac{1}{2} \int_0^L dx [(\partial_t \phi)^2 + (\partial_x \phi)^2]$$

and

$$\langle 0|a_m a_n^\dagger|0\rangle = \delta_{mn}; \quad \text{others} = 0$$

We obtain

$$\varepsilon_0 \equiv \frac{1}{L} \langle 0|H|0\rangle = \frac{\pi}{2L^2} \sum_{n=1}^{\infty} n \sim \text{diverges}$$

As $L \rightarrow \infty$ this diverges and

$$\varepsilon_0^{\text{free}} = \lim_{L \rightarrow \infty} \varepsilon_0(L)$$

This energy should not be infinite and should be ~ 0 !² We are led to

$$\Delta\varepsilon(L) = \varepsilon_0(L) - \varepsilon_0^{\text{free}} \sim \infty - \infty$$

Which is absurd.

We regulate ε_0 and $\varepsilon_0^{\text{free}}$ by a cutoff hoping that the physical observables we compute become finite and independent of the cutoff. We introduce an exponential cutoff

$$\varepsilon_0(L, \alpha) = \frac{\pi}{2L^2} \sum_{n=1}^{\infty} n \exp\left(-\frac{n\alpha}{L}\right)$$

with α being the cutoff parameter. We can write

$$\begin{aligned} \varepsilon_0(L, \alpha) &= -\frac{\pi}{2L} \frac{\partial}{\partial \alpha} \sum_{n=1}^{\infty} \exp\left(-\frac{n\alpha}{L}\right) \\ &= \frac{\pi}{8L^2} \sinh^{-2}\left(\frac{\alpha}{2L}\right) \end{aligned}$$

Taking the limit $\alpha \rightarrow 0$ where the cutoff vanishes we then have

$$\varepsilon_0(L, \alpha) = \underbrace{\frac{\pi}{2\alpha^2}}_{\sim \text{divergent}} - \frac{\pi}{24L^2} + \underbrace{\mathcal{O}(\alpha^2)}_{\text{vanishes}}$$

Consider now $\Delta\varepsilon(L) = \varepsilon_0(L) - \varepsilon_0^{\text{free}}$ then

$$\begin{aligned} \Delta\varepsilon_{\text{renorm}}(L) &= \lim_{\alpha \rightarrow 0} \left[\varepsilon_0(L, \alpha) - \lim_{L \rightarrow \infty} \varepsilon_0(L, \alpha) \right] \\ &= \lim_{\alpha \rightarrow 0} \left(\frac{\pi}{2\alpha^2} - \frac{\pi}{24L^2} + \dots - \frac{\pi}{2\alpha^2} \right) \end{aligned}$$

²Assuming we ignore dark energy which is negligible.

$$= -\frac{\pi}{24L^2}$$

Then the zero-point energy in the presence of plates is lower by a finite amount as compared to the free case!

This vacuum energy is a form of potential energy depending on the distance L between the plates. Then by definition we have a force

$$\begin{aligned} F_{\text{Casimir}} &= -\frac{d}{dL} \Delta E \\ &= -\frac{d}{dL} (L \Delta \varepsilon_{\text{renorm}}) \\ &= -\frac{\pi}{24L^2} \end{aligned}$$

This force is negative meaning the plates get pulled together. The increased number of modes outside the plates gives rise to a pressure!

4.3. A note on the cosmological constant

To determine the physical vacuum energy in curved space we will follow a procedure similar to the above by subtracting the flat space vacuum energy. We will find that in the *Friedmann-Robertson-Walker* spacetime the vacuum energy of a scalar field is the same as in flat space. Then we expect the renormalised cosmological constant to vanish. However we do observe a tiny cosmological constant

$$\frac{\Lambda}{G_N} \sim M_p^2 H_0^2 \sim 10^{-120} M_p^4$$

This is much smaller than expected with a cutoff at the Planck scale $\sim M_p^4$ and non-zero. We might expect this vacuum energy has some dynamical origin which we refer to as *dark energy*.

Removing the UV divergence as above can be interpreted physically as not knowing the theory above the UV cutoff. We remove some negative contribution to the vacuum energy

given by the unknown physics above the cutoff. This is a problem since the cosmological constant becomes very sensitive to these unknown UV physics. We can write

$$\Lambda_{\text{renorm}} = \Lambda_{\text{bare}} + \Lambda_{\text{counter-term}}$$

where $\Lambda_{\text{bare}} \sim M_p^4$ and $\Lambda_{\text{counter-term}}$ is some counter term which depends on the UV physics. This counter term is supposed to cancel Λ_{bare} with a precision on the order of 10^{120} . Then any small change in the UV physics such as a phase transition, symmetry breaking etc. will change the IR cosmological constant enormously! This *fine-tuning* seems absurd and is typically referred to as the *cosmological constant problem*.

As a concrete example consider

$$\Lambda_{\text{renorm}}^{\text{TeV}} = \Lambda_{\text{bare}}^{\text{TeV}} + \Lambda_{\text{CT}}^{\text{TeV}}$$

When ~ 100 GeV we have the electroweak phase transition which changes the vacuum energy by some significant ΔV meaning

$$\Lambda_{\text{obs}} \rightarrow \Lambda_{\text{obs}}^{\text{TeV}} + \Delta V$$

This breaks our previous fine-tuning. We could instead consider

$$\Lambda_{\text{renorm}}^{\text{meV}} = \Lambda_{\text{bare}}^{\text{meV}} + \Lambda_{\text{CT}}^{\text{meV}}$$

When $\sim \text{meV}$ all *fundamental* phase transitions have occurred so nothing in principal should break our fine-tuning. However now $\Lambda_{\text{bare}}^{\text{meV}}$ is implied to know everything about what has happened at larger scales while also being very sensitive to what happened. This is quite non-sensical and runs in to many problems. We strongly believe that scales do decouple. The minute details of e.g. quantum gravity should not change the dynamics of relativity at very large scales.

5. Fields in Curved Space

5.1. The scalar field

We consider a scalar field in curved space. The Lagrangian for such a field is

$$\mathcal{L} = -\frac{1}{2}\sqrt{-g}[g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi + (m^2 + \xi R)\phi^2]$$

with ξ being the *non-minimal coupling*. We have the metric

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu; \quad \mu, \nu = 0, 1, \dots, d-1$$

and we define

$$g \equiv \det g_{\mu\nu}$$

The term ξR is the only possible local scalar coupling in $\mathcal{L}$ by dimensional analysis. The inclusion of other, higher dimensional couplings would require the introduction of some mass scale.

By varying the action

$$S = \int d^n x \mathcal{L}$$

we trivially find the equation of motion³

$$(\square + m^2 + \xi R)\phi = 0$$

and we can write⁴

$$\begin{aligned} \square\phi &\equiv g^{\mu\nu}\nabla_\mu\nabla_\nu\phi \\ &= \frac{1}{\sqrt{-g}}\partial_\mu(\sqrt{-g}g^{\mu\nu}\partial_\nu\phi) \end{aligned}$$

We say the field is *minimally coupled* since when $\xi \rightarrow 0$ we recover flat space. When

$$\xi = \frac{1}{4} \frac{n-2}{n-1} \stackrel{n=4}{\sim} \frac{1}{6}$$

The field has *conformal coupling* since the equations of motion with $m = 0$ become invariant under *conformal transformations* $g_{\mu\nu} \rightarrow \Omega^2 g_{\mu\nu}$.

³This is simply the Klein-Gordon equation with the mass m shifted.

⁴Using $(-, +, +, +)$.

We seek an orthonormal basis for the solutions to the above equation. This means we need a covariant definition of the inner product. We define

$$(\phi_1, \phi_2) = -i \int_{\Sigma} \phi_1(\mathbf{x}) \vec{\nabla}_{\mu} \phi_2^*(\mathbf{x}) \sqrt{-g_{\Sigma}} d\Sigma^{\mu}$$

with $d\Sigma^{\mu} = n^{\mu} d\Sigma$ where n^{μ} is the *future-directed* unit vector orthogonal to the *spacelike* hypersurface Σ and

$$\phi_1 \vec{\nabla}_{\mu} \phi_2^* \equiv \phi_1 \nabla_{\mu} \phi_2^* - \phi_2^* \nabla_{\mu} \phi_1 \sim j_{\mu}$$

Consider

$$\begin{aligned} D_{\mu} j^{\mu} &= \nabla_{\mu} (\phi_1 \nabla_{\mu} \phi_2^* - \phi_2^* \nabla_{\mu} \phi_1) \\ &= \phi_1 \square \phi_2^* - \phi_2^* \square \phi_1 \\ &= \phi_1 (-m^2 - \xi R) \phi_2^* - \phi_2^* (-m^2 - \xi R) \phi_1 \\ &= 0 \end{aligned}$$

meaning j_{μ} is conserved. We now take two spacelike hypersurfaces Σ_1 and Σ_2 that bound a spacetime region $\mathcal{V}$. Then by Gauss' theorem

$$\int_{\mathcal{V}} d^d x \sqrt{-g} D_{\mu} j^{\mu} = \int_{\partial \mathcal{V}} d\Sigma_{\mu} j^{\mu}$$

The LHS vanishes so

$$0 = \int_{\Sigma_2} d\Sigma_{\mu} j^{\mu} - \int_{\Sigma_1} d\Sigma_{\mu} j^{\mu} + \underbrace{\text{boundary terms}}_{\rightarrow 0}$$

implying

$$\int_{\Sigma_1} d\Sigma_{\mu} j^{\mu} = \int_{\Sigma_2} d\Sigma_{\mu} j^{\mu} \sim Q(\Sigma)$$

meaning (ϕ_1, ϕ_2) is independent of the choice of Σ . With Minkowski spacetime we recover the usual *Klein-Gordon inner product*

$$(\phi_1, \phi_2) \stackrel{\text{flat}}{=} -i \int_t \phi_1 \vec{\partial}_t \phi_2^* d^3 \mathbf{x}$$

The orthonormal modes satisfy

$$(u_i, u_j) = \delta_{ij}; \quad (u_i^*, u_j^*) = -\delta_{ij}; \quad (u_i, u_j^*) = 0$$

which are called the *Wronskian conditions*. Consider the Minkowski modes

$$u_{\mathbf{k}} \sim e^{\pm i\mathbf{k}\cdot\mathbf{x} \mp i\omega t}$$

These satisfy the Wronskian conditions meaning the above definition is a valid covariant generalisation of the inner product.⁵

Then some general solution can be written as

$$\phi(\mathbf{x}) = \sum_i (a_i u_i(\mathbf{x}) + a_i^\dagger u_i^*(\mathbf{x}))$$

Quantizing the field is done as before by imposing the commutation relations

$$[a_i, a_j^\dagger] = \delta_{ij}; \quad \text{others} = 0$$

5.2. Bogolubov transformations

The Minkowski metric is time-independent implying the existence of a $\frac{\partial}{\partial t}$ *Killing vector* orthonormal to the spacelike hypersurface where t is constant. The Minkowski modes are eigenfunctions of $\frac{\partial}{\partial t}$ with eigenvalues $-i\omega$. We have something similar for spatial translations and boosts and the Minkowski vacuum is the state invariant under the Poincaré group. This defines a *unique* vacuum. However in curved space Poincaré invariance breaks meaning there is no unique vacuum in general. The existence of Killing vectors can amend this.⁶

We want to compare two different orthonormal sets u_i and $\bar{u}_j$ and the resulting equivalent expansions

$$\phi = \sum_i (a_i u_i + a_i^\dagger u_i^*); \quad \phi = \sum_j (\bar{a}_j \bar{u}_j + \bar{a}_j^\dagger \bar{u}_j^*)$$

We are lead to two definitions of the vacuum

$$a_i |0\rangle = 0; \quad \bar{a}_j |\bar{0}\rangle = 0$$

⁵By the *principle of general covariance*.

⁶We can view this as general covariance making coordinates physically meaningless. This implies no unique mode decomposition exists and as a consequence there is no well-defined frequency or conserved energy.

The bases are assumed to be complete implying we can write

$$\bar{u}_j = \sum_i (\alpha_{ji} u_i + \beta_{ji} u_i^*); \quad u_i = \sum_j (\alpha_{ji}^* \bar{u}_j - \beta_{ji} \bar{u}_j^*)$$

These are called the *Bogolubov transformations* with α_{ij} and β_{ij} being the *Bogolubov coefficients*. We immediately find

$$\alpha_{ij} = (\bar{u}_i, u_j); \quad \beta_{ij} = -(\bar{u}_i, u_j^*)$$

Then we have

$$\sum_i (a_i u_i + a_i^\dagger u_i^*) = \sum_i (\bar{a}_i \bar{u}_i + \bar{a}_i^\dagger \bar{u}_i^*)$$

Taking $(\dots, u_i)$ we obtain

$$a_i = \sum_j [\bar{a}_j (\bar{u}_j, u_i) + \bar{a}_j^\dagger (\bar{u}_j^*, u_i)] = \sum_j (\alpha_{ji} \bar{a}_j + \beta_{ji}^* \bar{a}_j^\dagger)$$

similarly

$$\bar{a}_j = \sum_i (\alpha_{ji} a_i + \beta_{ji}^* a_i^\dagger)$$

We see that for $\beta_{ij} \neq 0$

$$a_i |\bar{0}\rangle = \sum_j \beta_{ji}^* |\bar{1}_j\rangle \neq 0$$

meaning the vacua are *truly different*. We also have for $N_i = a_i^\dagger a_i$

$$\langle \bar{0} | N_i | \bar{0} \rangle = \sum_j |\beta_{ji}|^2$$

implying the vacuum $|\bar{0}\rangle$ contains particles of the u_i mode!⁷

5.3. Killing vectors

We consider an integral curve $x^\mu(t)$ and define the vector field

$$V^\mu(x) \equiv \left. \frac{dx^\mu(t)}{dt} \right|_x \sim \text{tangents}$$

⁷This has many consequences. One such example is structure formation due to particle production when the vacuum changes with time in an expanding Universe. Another example is the *Unruh effect* where an accelerated observer observes particles in the Minkowski vacuum. We will discuss these in due time.

The *Lie derivative* $\mathcal{L}_V$ measures how any field changes as we drag it along the flow generated by V^μ . We simply define⁸

$$\mathcal{L}_V f = V^\mu \partial_\mu f$$

$$\mathcal{L}_V U^\mu = V^\nu \partial_\nu U^\mu - U^\nu \partial_\nu V^\mu$$

The Lie derivative of $g_{\mu\nu}$ is

$$\mathcal{L}_V g_{\mu\nu} = \nabla_\mu V_\nu + \nabla_\nu V_\mu$$

To see this consider a displacement along the direction given by some vector X^μ

$$x'^\mu = x^\mu + \varepsilon X^\mu$$

Then

$$\begin{aligned} g_{\mu\nu} &= (\delta_\mu^\rho + \varepsilon \partial_\mu X^\rho)(\delta_\nu^\sigma + \varepsilon \partial_\nu X^\sigma) \underbrace{g_{\rho\sigma}(x^\lambda + \varepsilon X^\lambda)}_{g_{\rho\sigma}(x) + \varepsilon X^\lambda \partial_\lambda g_{\rho\sigma}(x) + \dots} \\ &= g_{\mu\nu}(x) + \varepsilon (g_{\mu\sigma} \partial_\nu X^\sigma + g_{\nu\sigma} \partial_\mu X^\sigma + X^\lambda \partial_\lambda g_{\mu\nu}) + \mathcal{O}(\varepsilon^2) \\ &= g_{\mu\nu} + \varepsilon \mathcal{L}_X g_{\mu\nu} \end{aligned}$$

We say $\xi^\mu \equiv X^\mu$ is a *Killing vector* if the above operation is an *isometry* meaning

$$\mathcal{L}_\xi g_{\mu\nu} = \nabla_\mu \xi_\nu + \nabla_\nu \xi_\mu = 0$$

which is the *Killing equation*. This implies conserved quantities since $g_{\mu\nu}$ is invariant along the *flow* generated by ξ_μ .

We can explicitly pick coordinates (y, x^1, x^2, x^3) where

$$\xi^\mu = (\partial_y)^\mu \sim \text{basis vector along } y$$

or $\xi^\mu = \delta_y^\mu$. This collapses the definition of the Lie derivative as

$$\mathcal{L}_\xi \rightarrow \partial_y$$

implying $\partial_y g_{\mu\nu} = 0$. Consider the case mentioned before where $y = t$ meaning $\xi = \partial_t$ and

$$\partial_t g_{\mu\nu} = 0 \sim \text{stationary spacetime}$$

which is nice. Also since $\nabla_\mu T^{\mu\nu} = 0$ we have

⁸See e.g. *Nakahara* for details.

$$\nabla_\mu(\xi_\nu T^{\mu\nu}) = \frac{1}{\sqrt{-g}}\partial_\mu(\sqrt{-g}\xi_\nu T^{\mu\nu}) = 0$$

meaning the current $j^\mu = \xi_\nu T^{\mu\nu}$ is conserved. Assuming the spacetime is stationary we

have $\xi^\mu = \delta_t^\mu$ implying

$$\nabla_\mu T^\mu_t = 0 \sim \text{local energy conservation}$$

Likewise $\xi = \partial_\phi$ gives angular momentum conservation.

6. The Friedmann-Robertson-Walker Spacetime

By *the cosmological principle* the Universe appears homogeneous and isotropic on large scales. We also measure the Universe as being *flat*.⁹ The solution to the Einstein field equations with these assumptions is the *Friedmann-Robertson-Walker metric*

$$ds^2 = -dt^2 + a^2(t) d\mathbf{x}^2$$

where $a(t)$ is the *scale factor*.

6.1. The scalar field

We assume the scalar field ϕ contributes a negligible energy density meaning. Then we can ignore its influence and treat $g_{\mu\nu}$ as fixed. We consider the free scalar field ϕ as before

$$S = -\frac{1}{2} \int d^4x \sqrt{-g} [g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi + m^2 \phi^2]$$

with the equation of motion

$$\square \phi + m^2 \phi = 0$$

We can simplify things by defining the *conformal time*

$$\eta(t) \equiv \int_{t_0}^t \frac{dt}{a(t)}, \quad a d\eta = dt$$

With η the metric becomes

$$ds^2 = a^2(\eta) [-d\eta^2 + d\mathbf{x}^2]$$

implying

$$g_{\mu\nu} = \underbrace{a^2(\eta)}_{\Omega^2} \eta_{\mu\nu} \sim \text{conformally flat}$$

We also make a *field redefinition*

$$\chi \equiv a(\eta) \phi$$

and trivially $\sqrt{-g} = a^4$. Then we find

$$S = -\frac{1}{2} \int d^3x d\eta \left[-(\partial_\eta \chi)^2 + (\nabla \chi)^2 + m_{\text{eff}}^2(\eta) \chi^2 \right]$$

⁹We measure $k = 0$.

with

$$m_{\text{eff}}^2(\eta) \equiv m^2 a^2 - \frac{1}{a} \partial_\eta^2 a$$

We see that *all* effects of gravity are within m_{eff} and S is time-dependent meaning the Hamiltonian will be time-dependent. This implies the energy of χ is not conserved.¹⁰

The equation of motion for χ becomes¹¹

$$\partial_\eta^2 \chi - \nabla^2 \chi + \left(m^2 a^2 - \frac{1}{a} \partial_\eta^2 a \right) \chi = 0$$

The ∇^2 is as in Minkowski space meaning we can expand in Fourier modes as

$$\chi(\eta, \mathbf{x}) = \int \frac{d^3 \mathbf{k}}{(2\pi)^{3/2}} \chi_{\mathbf{k}}(\eta) e^{i\mathbf{k} \cdot \mathbf{x}}$$

We then obtain

$$\partial_\eta^2 \chi_{\mathbf{k}} + \left(k^2 + m^2 a^2(\eta) - \frac{1}{a} \partial_\eta^2 a \right) \chi_{\mathbf{k}} = 0$$

Which can be written as

$$\partial_\eta^2 \chi_{\mathbf{k}} + \omega_{\mathbf{k}}^2(\eta) \chi_{\mathbf{k}} = 0$$

with $\omega_{\mathbf{k}}^2 \equiv k^2 + m_{\text{eff}}^2(\eta)$. We assume we know a function $V_{\mathbf{k}}(\eta)$ satisfying¹²

$$\partial_\eta^2 V_{\mathbf{k}} + \omega_{\mathbf{k}}^2 V_{\mathbf{k}} = 0$$

Then the general solution $\chi_{\mathbf{k}}(\eta)$ can be written as

$$\chi_{\mathbf{k}}(\eta) = \frac{1}{\sqrt{2}} (a_{\mathbf{k}}^- V_{\mathbf{k}}^* + b_{\mathbf{k}} V_{\mathbf{k}})$$

We assume χ is real implying $\chi_{\mathbf{k}}^* = \chi_{-\mathbf{k}}$. Then

$$(a_{\mathbf{k}}^-)^* \stackrel{!}{=} b_{-\mathbf{k}} \equiv a_{\mathbf{k}}^+$$

and we obtain

$$\chi_{\mathbf{k}}(\eta) = \frac{1}{\sqrt{2}} (a_{\mathbf{k}}^- V_{\mathbf{k}}^* + a_{-\mathbf{k}}^+ V_{\mathbf{k}})$$

¹⁰This leads to particle creation.

¹¹We use $a^2 \square \phi = -\partial_\eta^2 \phi - 2 \frac{1}{a} \partial_\eta a \partial_\eta \phi + \nabla^2 \phi$.

¹² $\omega_{\mathbf{k}}^2 = \omega_{-\mathbf{k}}^2$ implies $V_{\mathbf{k}} = V_{-\mathbf{k}}$.

Then

$$\begin{aligned}\chi(\eta, \mathbf{x}) &= \int \frac{d^3\mathbf{k}}{(2\pi)^{3/2}} \frac{1}{\sqrt{2}} (a_{\mathbf{k}}^- V_{\mathbf{k}}^* + a_{-\mathbf{k}}^+ V_{-\mathbf{k}}) e^{i\mathbf{k}\cdot\mathbf{x}} \\ &= \int \frac{d^3\mathbf{k}}{(2\pi)^{3/2}} \frac{1}{\sqrt{2}} (a_{\mathbf{k}}^- V_{\mathbf{k}}^* e^{i\mathbf{k}\cdot\mathbf{x}} + a_{\mathbf{k}}^+ V_{\mathbf{k}} e^{-i\mathbf{k}\cdot\mathbf{x}})\end{aligned}$$

We can show

$$\frac{a_{\mathbf{k}}^-}{\sqrt{2}} = \frac{\partial_\eta V_{\mathbf{k}} \chi_{\mathbf{k}} - V_{\mathbf{k}} \partial_\eta \chi_{\mathbf{k}}}{\partial_\eta V_{\mathbf{k}} V_{\mathbf{k}}^* - V_{\mathbf{k}} \partial_\eta V_{\mathbf{k}}^*} = \frac{W[V_{\mathbf{k}}, \chi_{\mathbf{k}}]}{W[V_{\mathbf{k}}, V_{\mathbf{k}}^*]}$$

6.2. Quantization

We quantize χ by promoting $a_{\mathbf{k}}^\pm$ to operators

$$a_{\mathbf{k}}^- \rightarrow a_{\mathbf{k}}; \quad a_{\mathbf{k}}^+ \rightarrow a_{\mathbf{k}}^\dagger$$

and obtain

$$\chi(\eta, \mathbf{x}) = \int \frac{d^3\mathbf{k}}{(2\pi)^{3/2}} \frac{1}{\sqrt{2}} (a_{\mathbf{k}} V_{\mathbf{k}}^* e^{i\mathbf{k}\cdot\mathbf{x}} + a_{\mathbf{k}}^\dagger V_{\mathbf{k}} e^{-i\mathbf{k}\cdot\mathbf{x}})$$

We impose the usual commutators

$$[a_{\mathbf{k}}, a_{\mathbf{k}'}^\dagger] = \delta(\mathbf{k} - \mathbf{k}'), \quad \text{others} = 0$$

implying¹³

$$[\chi(\eta, \mathbf{x}_1), \pi(\eta, \mathbf{x}_2)] = i\delta(\mathbf{x}_1 - \mathbf{x}_2)$$

when

$$\text{Im } V_{\mathbf{k}}^* \partial_\eta V_{\mathbf{k}} = W[V_{\mathbf{k}}, V_{\mathbf{k}}^*] = 2i$$

This ensures the Wronskian conditions are satisfied by the modes $u_{\mathbf{k}}$ ¹⁴

$$u_{\mathbf{k}}(\eta, \mathbf{x}) = \frac{1}{(2\pi)^{3/2}} \frac{1}{\sqrt{2}} V_{\mathbf{k}}(\eta) e^{i\mathbf{k}\cdot\mathbf{x}}$$

We now need to *choose* a vacuum. This choice is not unique since above $a_{\mathbf{k}} \sim V_{\mathbf{k}}$. The vacuum is typically chosen by imposing some physical constraint on the system.¹⁵

¹³Where $\pi(\eta, \mathbf{x}) = \frac{\partial \mathcal{L}}{\partial(\partial_\eta \chi)} = \partial_\eta \chi(\eta, \mathbf{x})$.

¹⁴This normalizes our $V_{\mathbf{k}}$.

¹⁵This usually depends on which observer we are trying to describe.

6.3. The instantaneous vacuum

We attempt to define vacuum as the smallest energy eigenstate at some time $\eta = \eta_0$ of the *instantaneous* Hamiltonian. The Hamiltonian of χ is

$$H = \frac{1}{2}\pi^2 + \frac{1}{2}\omega^2(\eta)\chi^2$$

We can obtain

$$\begin{aligned} E(\eta) &\equiv \langle 0|H(\eta)|0\rangle \\ &= \int \frac{d^3\mathbf{k}}{(2\pi)^3} \frac{|\partial_\eta V_{\mathbf{k}}|^2 + \omega_{\mathbf{k}}^2 |V_{\mathbf{k}}|^2}{4} \end{aligned}$$

This is minimal at $\eta = \eta_0$ when

$$|\partial_\eta V_{\mathbf{k}}|^2 + \omega_{\mathbf{k}}^2 |V_{\mathbf{k}}|^2$$

is minimal for all k -modes at $\eta = \eta_0$. We define

$$s_{\mathbf{k}} = V_{\mathbf{k}}(\eta_0); \quad t_{\mathbf{k}} = \partial_\eta V_{\mathbf{k}}(\eta_0)$$

with the Wronskian condition

$$s_{\mathbf{k}}^* t_{\mathbf{k}} - s_{\mathbf{k}} t_{\mathbf{k}}^* = 2i$$

We now want to minimize

$$|t_{\mathbf{k}}|^2 + \omega^2(\eta_0) |s_{\mathbf{k}}|^2$$

Taking $s_{\mathbf{k}}$ to be real and writing $t_{\mathbf{k}} = t_{1\mathbf{k}} + it_{2\mathbf{k}}$ we obtain

$$s_{\mathbf{k}} = \frac{1}{t_{2\mathbf{k}}}$$

Then

$$4E(\eta_0) = \int \frac{d^3\mathbf{k}}{(2\pi)^3} \left(t_{1\mathbf{k}}^2 + t_{2\mathbf{k}}^2 + \frac{\omega_{\mathbf{k}}^2(\eta_0)}{t_{2\mathbf{k}}^2} \right)$$

When $\omega_{\mathbf{k}}^2(\eta_0) > 0$ we have a minima for

$$t_{1\mathbf{k}} = 0; \quad t_{2\mathbf{k}} = \sqrt{\omega_{\mathbf{k}}(\eta_0)}$$

When $\omega_{\mathbf{k}}^2(\eta_0) < 0$ no minima exists. Then for $\omega_{\mathbf{k}}^2(\eta_0) > 0$ the *instantaneous vacuum* is given by

$$V_{\mathbf{k}}(\eta_0) = \frac{1}{\sqrt{\omega_{\mathbf{k}}(\eta_0)}}; \quad \partial_{\eta} V_{\mathbf{k}}(\eta_0) = i\sqrt{\omega_{\mathbf{k}}(\eta_0)} = i\omega_{\mathbf{k}}(\eta_0)V_{\mathbf{k}}(\eta_0)$$

This agrees¹⁶ with the Minkowski spacetime result where $\omega_{\mathbf{k}}$ is time-independent yielding

$$V_{\mathbf{k}}(\eta) = \frac{1}{\sqrt{\omega_{\mathbf{k}}}} e^{-i\omega_{\mathbf{k}}\eta}$$

When we introduce a time-dependent background this vacuum changes over time¹⁷

$$|0\rangle_{\eta_0} \neq |0\rangle_{\eta_1}$$

Consider two sets of modes $V_{\mathbf{k}}(\eta)$ and $U_{\mathbf{k}}(\eta)$ describing the vacuum at $\eta = \eta_0$ and $\eta = \eta_1$ respectively and assume they are both known at $\eta = \eta_0$. We can relate them using the Bogolubov coefficients $\alpha_{\mathbf{k}}$ and $\beta_{\mathbf{k}}$

$$V_{\mathbf{k}}^* = \alpha_{\mathbf{k}} U_{\mathbf{k}}^* + \beta_{\mathbf{k}} U_{\mathbf{k}}; \quad \partial_{\eta} V_{\mathbf{k}}^* = \alpha_{\mathbf{k}} \partial_{\eta} U_{\mathbf{k}}^* + \beta_{\mathbf{k}} \partial_{\eta} U_{\mathbf{k}}$$

where

$$\alpha_{\mathbf{k}} = \left. \frac{\partial_{\eta} U_{\mathbf{k}} V_{\mathbf{k}}^* - U_{\mathbf{k}} \partial_{\eta} V_{\mathbf{k}}^*}{2i} \right|_{\eta_0} \quad \beta_{\mathbf{k}}^* = \left. \frac{\partial_{\eta} U_{\mathbf{k}} V_{\mathbf{k}} - U_{\mathbf{k}} \partial_{\eta} V_{\mathbf{k}}}{2i} \right|_{\eta_0}$$

We could have used $\eta \rightarrow -\infty$ instead of $\eta = \eta_0$. All states with finite η_0 would then appear excited. We would have particle creation as the Universe evolved ex nihilo!

6.4. The adiabatic vacuum

When $\omega_{\mathbf{k}}$ varies slowly with η the *adiabatic vacuum* is useful. We still have

$$\partial_{\eta}^2 V_{\mathbf{k}} + \omega_{\mathbf{k}}^2 V_{\mathbf{k}} = 0, \quad \omega_{\mathbf{k}}^2 = k^2 + m_{\text{eff}}^2$$

This has a formal *WKB-type* solution

$$V_{\mathbf{k}} = \frac{1}{\sqrt{W_{\mathbf{k}}}} \exp \left[-i \int^{\eta} W_{\mathbf{k}}(\eta') d\eta' \right]$$

with¹⁸

$$W_{\mathbf{k}}^2(\eta) = \omega_{\mathbf{k}}^2(\eta) - \frac{1}{2} \left[\frac{1}{W_{\mathbf{k}}} \partial_{\eta}^2 W_{\mathbf{k}} - \frac{3}{2} \frac{1}{W_{\mathbf{k}}^2} (\partial_{\eta} W_{\mathbf{k}})^2 \right]$$

¹⁶Up to a phase.

¹⁷With each instantaneous vacuum mimicking the Minkowski vacuum.

¹⁸This is our *dispersion relation*.

The *adiabatic limit* has

$$W_{\mathbf{k}} \simeq W_{\mathbf{k}}^{(0)}(\eta) = \omega_{\mathbf{k}}(\eta)$$

and we obtain

$$V_{\mathbf{k}}^{(0)} = \frac{1}{\sqrt{\omega_{\mathbf{k}}}} \exp \left[-i \int_{\eta_0}^{\eta} \omega_{\mathbf{k}}(\eta') d\eta' \right]$$

This solution approximately satisfies the instantaneous minimum energy condition. We have

$$V_{\mathbf{k}}(\eta_0) = V_{\mathbf{k}}^{(0)}(\eta_0); \quad \partial_{\eta} V_{\mathbf{k}}(\eta) \Big|_{\eta_0} = \partial_{\eta} V_{\mathbf{k}}^{(0)}(\eta) \Big|_{\eta_0}$$

implying

$$V_{\mathbf{k}}(\eta_0) = \frac{1}{\sqrt{\omega_{\mathbf{k}}(\eta_0)}}; \quad \partial_{\eta} V_{\mathbf{k}}(\eta) \Big|_{\eta_0} = \left[-i\omega_{\mathbf{k}} - \frac{1}{2\omega_{\mathbf{k}}} \partial_{\eta} \omega_{\mathbf{k}} \right] \frac{1}{\sqrt{\omega_{\mathbf{k}}}} \Big|_{\eta_0}$$

Then the energy becomes

$$\begin{aligned} E_{\mathbf{k}} &= \frac{1}{4} \left(|\partial_{\eta} V_{\mathbf{k}}|^2 + \omega_{\mathbf{k}}^2 |V_{\mathbf{k}}|^2 \right) \\ &= \frac{1}{2} \omega_{\mathbf{k}} + \frac{1}{16} \frac{1}{\omega_{\mathbf{k}}^3} (\partial_{\eta} \omega_{\mathbf{k}})^2 \\ &\simeq \frac{1}{2} \omega_{\mathbf{k}} \end{aligned}$$

which is the same as the instantaneous vacuum for $\eta = \eta_0$.

We similarly define the second order adiabatic vacuum using

$$\left(W_{\mathbf{k}}^{(2)} \right)^2 = \omega_{\mathbf{k}}^2 - \frac{1}{2} \left[\frac{1}{\omega_{\mathbf{k}}} \partial_{\eta}^2 \omega_{\mathbf{k}} - \frac{3}{2} \frac{1}{\omega_{\mathbf{k}}^2} (\partial_{\eta} \omega_{\mathbf{k}})^2 \right]$$

This will differ from the instantaneous vacuum.

7. The FRW cosmology

The FRW metric can be written as

$$ds^2 = -dt^2 + a^2(t) \left[\frac{dr^2}{1 - kr^2} + r^2 d\Omega^2 \right]$$

and we can treat the Universe as a co-moving ideal fluid with

$$T_{\mu\nu} = \text{diag}(\rho, p, p, p)$$

Then from the EFE we find the *Friedmann equations*

$$H^2 = \frac{8\pi G_N}{3} \rho - \frac{k}{a^2}$$

$$\frac{\ddot{a}}{a} = -\frac{4\pi G_N}{3} (\rho + 3p)$$

with H being the *Hubble parameter*

$$H \equiv \frac{\dot{a}}{a}$$

We also have

$$D^\nu T_{\mu\nu} = 0$$

implying

$$\dot{\rho} + 3H(\rho + p) = 0$$

which is simply the *continuity equation*.¹⁹ We usually assume an equation of state of the form $p = w\rho$ where w is constant. Then

$$\rho \sim a^{-3(w+1)}$$

which in a spatially flat Universe implies

$$a \sim t^{2/3(1+w)}$$

We care about specific values of w ,

$$w_\Lambda = -1 \Rightarrow \rho_\Lambda \sim \text{constant}$$

$$w_r = \frac{1}{3} \Rightarrow \rho_r \sim a^{-4}$$

$$w_m = 0 \Rightarrow \rho_m \sim a^{-3}$$

¹⁹The continuity equation is not independent and follows directly from the Friedmann equations.

By comparing the scaling of the components we see the Universe will experience *eras* of domination with certain components dominating the rest.

We can quantify the amount of curvature by defining the *critical* energy density

$$\rho_c = \frac{3H^2}{8\pi G_N}$$

This is the energy density we would measure if the Universe were spatially flat. Then we can define *density parameters*

$$\Omega = \frac{\rho}{\rho_c}$$

implying

$$\Omega - 1 = \frac{k}{a^2 H^2}$$

Then Ω fully determines the spatial curvature k .

7.1. The Big Bang

The Friedmann equations and measurements imply

$$a(t = t_{\text{BB}}) \rightarrow 0$$

with $t_{\text{BB}} \sim 0$ being some finite time in the past. We also have

$$\rho_r \rightarrow \infty$$

as $a \rightarrow 0$, implying the Universe expanded from a hot and dense radiation-dominated state. This is called the *Big Bang model*. By assumption photons originating from the Big Bang can only have travelled a finite distance. This implies the *observable* Universe is finite, however, the *true* Universe may be infinite.

We can make the notion of *size* precise by defining the *particle horizon*. We consider photons moving radially and satisfying²⁰

$$dr = \frac{dt}{a(t)}$$

²⁰We have $d_H \sim H^{-1}$ when $a \sim t^\beta$ for $\beta < 1$.

The particle horizon is then the *physical* distance photons have travelled since t_{BB} at some time t

$$d_H(t) = a(t) \int_0^t \frac{dt'}{a(t')}$$

When the Universe was radiation-dominated we had $a \sim t^{-1/2}$ implying

$$d_H(t) = \frac{1}{H}$$

Then any physical scale λ outside the horizon

$$\lambda \gg \frac{1}{H}$$

is not in *causal contact*. We call H^{-1} the *Hubble radius*.

When

$$t_{\text{rec}} \sim 380000 \text{ years}$$

the Universe had cooled sufficiently due to the expansion that electrons and protons could form neutral hydrogen

$$e^- + p \rightarrow H$$

Then after t_{rec} the Universe becomes transparent to photons leaving us with the *cosmic microwave background radiation*. The CMB is a blackbody at the temperature

$$T_{\text{CMB}} \sim 2.735 \text{ K}$$

with inhomogeneities on the order

$$\frac{\Delta T}{T} \sim 10^{-4}$$

This is a problem.

7.2. The horizon problem

To see why consider the size of the observable Universe at t_{LS} ²¹

$$\lambda_H(t_{\text{LS}}) = d_H(t_0) \left(\frac{a_{\text{LS}}}{a_0} \right) = d_H(t_0) \left(\frac{T_0}{T_{\text{LS}}} \right)$$

²¹This is the largest physical scale λ .

We compare this with the Hubble radius. Assuming the Universe was matter-dominated at t_{LS} then

$$H^2 \sim T^3$$

implying

$$H_{\text{LS}}^{-1} = d_H(t_{\text{LS}}) = d_H(t_0) \left(\frac{T_{\text{LS}}}{T_0} \right)^{-3/2}$$

Then

$$\frac{\lambda_H^3}{d_H^3} \sim 10^6$$

We find there were $\sim 10^6$ causally disconnected *Hubble volumes* within the volume corresponding to the observable Universe at t_{LS} . This is weird since we observe the CMB to be homogeneous. This is called the *horizon problem*.

The reason the above is a problem is d_H being finite. This happens since $\ddot{a} < 0$ implying the lower limit of d_H converges. The problem would then be solved if we had $\ddot{a} > 0$ implying the lower limit of d_H diverges. Then a phase just after t_{BB} and before radiation-domination with $\ddot{a} > 0$ would *push* out d_H allowing the Universe to *thermalize* thereby solving the horizon problem as in Figure 1. This period is called *cosmic inflation*.

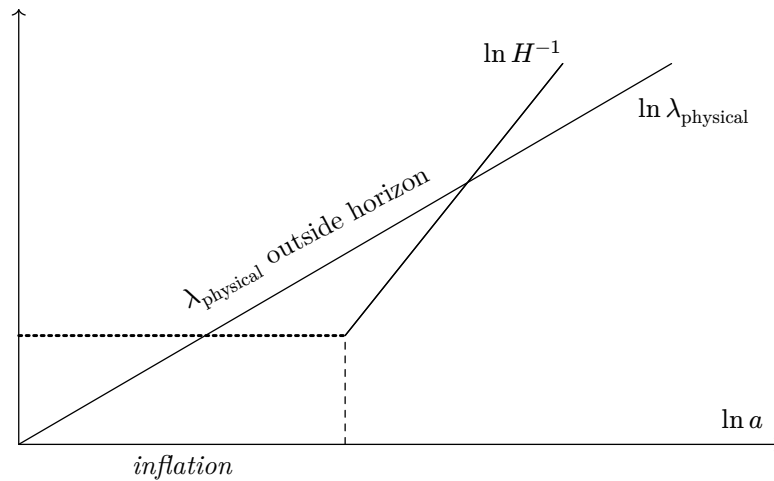


Figure 1: How inflation solves the horizon problem.

Consider

$$\frac{\ddot{a}}{a} = -\frac{4\pi G_N}{3}(\rho + 3p)$$

we find $\ddot{a} > 0$ implies

$$p < -\frac{\rho}{3}$$

This would occur if $p = -\rho$ implying²²

$$\rho = \text{constant}$$

Then

$$H^2 = \frac{8\pi G_N}{3}\rho$$

implies

$$H \simeq \text{constant}$$

during inflation. We find

$$a \sim e^{H_{\text{inf}} t}$$

which is called *de Sitter spacetime*.

8. Inflation

To describe inflation we seek a field theoretic description of a fluid with

$$p < -\frac{\rho}{3}$$

This is simplest by considering a scalar field ϕ referred to as the *inflaton*. The action describing such a field is²³

$$\begin{aligned} S &= \int d^4x \sqrt{-g} \mathcal{L} \\ &= \int dt d^3x a^3 \left[\frac{1}{2} \dot{\phi}^2 - \frac{1}{2} \partial_i \phi \partial^i \phi - V(\phi) \right] \end{aligned}$$

Using the Euler-Lagrange equations we find

²²This could be through a cosmological constant Λ .

²³Assuming minimal coupling.

$$\ddot{\phi} + 3H\dot{\phi} - \frac{1}{a^2}\nabla^2\phi + V_\phi(\phi) = 0$$

with

$$V_\phi = \frac{dV}{d\phi}$$

The energy-momentum tensor is

$$\begin{aligned} T_{\mu\nu} &= -\frac{2}{\sqrt{-g}} \frac{\delta S}{\delta g^{\mu\nu}} \\ &= \partial_\mu\phi\partial_\nu\phi + g_{\mu\nu}\mathcal{L} \end{aligned}$$

We compare this with the energy-momentum tensor of an ideal fluid

$$T_{\mu\nu} = \begin{pmatrix} \rho & & & \\ & a^2p & & \\ & & a^2p & \\ & & & a^2p \end{pmatrix}$$

Then by inspection

$$\begin{aligned} T_{00} &= \rho_\phi \\ &= \frac{1}{2}\dot{\phi}^2 + V(\phi) + \frac{1}{2a^2}(\nabla\phi)^2 \\ \frac{T_{ii}}{a^2} &= p_\phi \\ &= \frac{1}{2}\dot{\phi}^2 - V(\phi) - \frac{1}{6a^2}(\nabla\phi)^2 \end{aligned}$$

We see if the $\nabla\phi$ term dominated then

$$p_\phi = -\frac{\rho_\phi}{3}$$

which is insufficient to drive inflation. Alternatively, if the $\dot{\phi}^2$ term dominated then

$$p_\phi = \rho_\phi$$

which is obviously also insufficient. However, if the $V(\phi)$ term dominates then

$$p_\phi = -\rho_\phi$$

as required. We then want to describe inflation by a homogeneous $V(\phi)$ dominated scalar field. Therefore we expand ϕ as a perturbation on some homogeneous classical background

$$\phi(\mathbf{x}, t) = \phi_c(t) + \delta\phi(\mathbf{x}, t)$$

with

$$\langle 0 | \phi(\mathbf{x}, t) | 0 \rangle = \phi_c(t)$$

By assumption $\delta\phi$ is small so

$$\rho_{\phi_c} = \frac{1}{2}\dot{\phi}_c^2 + V(\phi_c)$$

$$p_{\phi_c} = \frac{1}{2}\dot{\phi}_c^2 - V(\phi_c)$$

Then if $V(\phi_c) \gg \dot{\phi}_c^2$ we have $p_{\phi_c} \simeq -\rho_{\phi_c}$.

8.1. Slow-roll inflation

The background is described by

$$\ddot{\phi}_c + 3H\dot{\phi}_c + V_\phi(\phi_c) = 0$$

Assuming $\dot{\phi}^2 \ll V(\phi)$ we need to also assume $\ddot{\phi}$ is small, otherwise inflation would stop quickly. These two assumptions are referred to as *slow-roll inflation*. Then we find the *slow-roll equations*

$$3H\dot{\phi}_c \simeq -V_\phi(\phi_c)$$

$$H^2 \simeq \frac{8\pi G_N}{3} V(\phi_c)$$

These are valid when the *slow-roll conditions*

$$\dot{\phi}_c^2 \ll V(\phi_c)$$

$$\ddot{\phi}_c \ll 3H\dot{\phi}_c$$

are satisfied. These imply

$$\frac{V_\phi^2}{V} \ll H^2; \quad V_{\phi\phi} \ll H^2$$

We define the *slow-roll parameters*

$$\epsilon_\phi = -\frac{\dot{H}}{H^2} = \frac{1}{16\pi G_N} \left(\frac{V_\phi}{V} \right)^2$$

$$\eta_\phi = \frac{1}{8\pi G_N} \frac{V_{\phi\phi}}{V} = \frac{V_{\phi\phi}}{3H^2}$$

Then the slow-roll conditions become²⁴

$$\epsilon_\phi \ll 1; \quad \eta_\phi \ll 1$$

We quantify the amount of inflation by the #e-foldings²⁵

$$N = \ln \frac{a_f}{a_i} = \int_{t_i}^{t_f} H dt$$

Usually we need $N \sim 60$.

There are many $V(\phi)$ satisfying the slow-roll conditions. We consider two broad classes of inflation: *large field models* and *small field models*. The large field models are described by a potential of the form

$$V(\phi) = \Lambda^4 \left(\frac{\phi}{\mu} \right)^p$$

with $p > 0$. Then

$$\eta_\phi = M_p^2 \frac{V_{\phi\phi}}{V} = p(p-1) \frac{M_p^2}{\phi^2}$$

and $\eta_\phi \ll 1$ implies $\phi > M_p$ for $p \neq 1$. Small field models are described by a potential of the form²⁶

$$V(\phi) = \lambda^4 \left[1 - \left(\frac{\phi}{\mu} \right)^p \right]$$

with $p > 0$. Then

²⁴We formally have inflation whenever $\epsilon_\phi < 1$.

²⁵From $a_f = a_i e^N$ since $a \sim e^{Ht}$ during inflation.

²⁶We can treat large field models as being models where there is nothing preventing ϕ from reaching the minima, while small fields models can exhibit false vacua. Then the large fields models can be understood with the notion of *chaotic inflation* where distinct pockets of inflation could emerge in the early Universe due to inhomogeneities. However, we may expect operators with ϕ to very large orders become important, as they would be unsuppressed when $\phi > M_p$. The small field models are motivated by particle physics, but are generally more complicated and less understood.

$$\eta_\phi = M_p^2 \frac{V_{\phi\phi}}{V} = -p(p-1) \left(\frac{\phi}{\mu} \right)^p \frac{M_p^2}{\phi^2}$$

assuming $\mu \ll M_p$ then $\eta_\phi \ll 1$ implies

$$\phi \ll \left(\frac{\mu}{M_p} \right)^{2/(p-2)} \quad \mu \ll \mu$$

for $p > 2$.

9. Field Perturbations

One reason why inflation had success was due to it making the Universe $\sim$ uniform. However, due to the rapid expansion small fluctuations in the inflation field become stretched to cosmological scales. This naturally induces ripples in spacetime due to the EFE coupling fluctuations in the inflation field with the metric. These ripples eventually seed perturbations in the CMB and matter. Thereby laying the blueprint for structure formation as these over- and underdensities becomes unstable in the late Universe. The reason we exist can then be attributed to these quantum fluctuations in the very early Universe!²⁷

9.1. Quantum fluctuations of a scalar field in quasi de Sitter

We consider a scalar field $\phi(\mathbf{x}, t)$ on a time-dependent background. We split the field as above

$$\phi(\mathbf{x}, t) = \phi_c(t) + \delta\phi(\mathbf{x}, t)$$

We would now like to quantize $\delta\phi(\mathbf{x}, t)$. Note,

$$\delta\phi(\mathbf{x}, t) = \int \frac{d^3k}{(2\pi)^{3/2}} e^{i\mathbf{k}\cdot\mathbf{x}} \delta\phi_k(t)$$

We assume ϕ is *sub-dominant* allowing us to ignore any backreaction on the metric. The equation of motion for the field becomes to linear order

$$\ddot{\phi}_c + \delta\ddot{\phi} + 3H(\dot{\phi}_c + \delta\dot{\phi}_c) - \frac{\nabla^2}{a^2} \delta\phi + \underbrace{V_\phi + V_{\phi\phi} \delta\phi}_{\text{Taylor expansion}} = 0$$

We apply the background equation of motion

$$\ddot{\phi}_c + 3H\dot{\phi}_c + V_\phi = 0$$

and insert the Fourier transform to find

$$\delta\ddot{\phi}_k + 3H\delta\dot{\phi}_k + \frac{k^2}{a^2} \delta\phi_k + m^2 \delta\phi_k = 0$$

²⁷ Assuming inflation holds water.

with $m^2 \equiv V_{\phi\phi}$. We can rewrite this using conformal time

$$\eta = \int \frac{dt}{a} = \int \frac{da}{a^2 H}$$

Assuming *plain* de Sitter with $H \sim \text{constant}$ we have

$$\eta = -\frac{1}{aH}$$

With slow-roll inflation we can consider time scales of order one *Hubble time* $\Delta t = H^{-1}$

and perform a slow-roll expansion

$$\begin{aligned} \eta &= \int e^{-Ht} dt \\ &= \int \exp \left[- \left(H_* + \frac{\dot{H}_*}{H_*} \right) t \right] dt \\ &= \int \exp \left[- H_* \left(1 + \frac{\dot{H}_*}{H_*^2} \right) t \right] dt \\ &= \int e^{-H_*(1-\epsilon_*)t} dt \\ &= -\frac{1}{aH_*(1-\epsilon_*)} \end{aligned}$$

Then to leading order in the slow-roll expansion

$$\eta = -\frac{1}{aH(1-\epsilon)}$$

We also introduce the canonical field $\chi = a\phi$ giving

$$\frac{\partial^2 \delta\chi_k}{\partial\eta^2} + \left(k^2 + m^2 a^2 - \frac{1}{a} \frac{\partial^2 a}{\partial\eta^2} \right) \delta\chi_k = 0$$

This is the equation for a SHO with

$$M^2(\eta) \equiv m^2 a^2 - \frac{1}{a} \frac{\partial^2 a}{\partial\eta^2}$$

Note,

$$\frac{\ddot{a}}{a} = \dot{H} + H^2; \quad \epsilon = -\frac{\dot{H}}{H^2}$$

Then

$$\frac{\partial^2 \delta\chi_k}{\partial\eta^2} + (k^2 + M^2) \delta\chi_k = 0$$

with

$$M^2(\eta) = \frac{1}{\eta^2} \left(\frac{m^2}{H^2} - 2 - 3\epsilon \right) + \mathcal{O}(\epsilon^2)$$

We define ν_ϕ by

$$M^2 = -\frac{1}{\eta^2} \left(\nu_\phi^2 - \frac{1}{4} \right) \Rightarrow \nu_\phi^2 = \frac{9}{4} + 3\epsilon - \frac{m^2}{H^2}$$

Then the equation of motion becomes a *Bessel equation*

$$\frac{\partial^2 \delta\chi_k}{\partial\eta^2} + \left[k^2 - \frac{1}{\eta^2} \left(\nu_\phi^2 - \frac{1}{4} \right) \right] \delta\chi_k = 0$$

with solutions in terms of the *Hankel functions* for $\nu_\phi \in \mathbb{R}$

$$\delta\chi_k = \sqrt{-\eta} \left[A_k H_\nu^{(1)}(-k\eta) + B_k H_\nu^{(2)}(-k\eta) \right]$$

The Hankel functions are defined in terms of the Bessel functions of the first and second kind as

$$H_\nu^{(1)}(x) = J_\nu(x) + iY_\nu(x); \quad H_\nu^{(2)}(x) = J_\nu(x) - iY_\nu(x)$$

We can then quantize the fluctuation field by promoting $\delta\chi_k$ to an operator and expanding as

$$\delta\hat{\chi}_k = \delta\chi_k a_k + \delta\chi_k^* a_{-k}^\dagger$$

9.2. The $m = 0$ scalar field in pure de Sitter

Assuming the scalar field is massless and we are in pure de Sitter then $v_\phi^2 = \frac{9}{4}$ and

$$\delta\chi_k = A_k \left(1 + \frac{i}{k\eta} \right) e^{-ik\eta} + B_k \left(1 - \frac{i}{k\eta} \right) e^{ik\eta}$$

On sub-horizon scales $k \gg aH$ so $-\eta k \gg 1$ and

$$\delta\chi_k \simeq A_k e^{-ik\eta} + B_k e^{ik\eta}$$

We could take

$$A_k = \frac{1}{\sqrt{2k}}; \quad B_k = 0$$

and recover the Minkowski vacuum in the infinite past

$$\delta\chi_k = \frac{1}{\sqrt{2k}} e^{-ik\eta}; \quad \eta \rightarrow -\infty$$

This is also the 0th order adiabatic vacuum, and we recover the usual dispersion relation

$$\omega_k^2(\eta) = k^2 + M^2 = k^2 - \frac{2}{\eta^2} \rightarrow k^2$$

Note, the instantaneous vacuum only exists for $\omega_k^2 > 0$

$$k^2 - \frac{2}{\eta^2} > 0$$

meaning only on sub-horizon scales with $(k\eta)^2 \gtrsim 2$ can the energy of a mode be normalized.²⁸ The instantaneous vacuum was determined by

$$V_k(\eta_0) = \frac{1}{\sqrt{2\omega_k(\eta_0)}}; \quad \frac{1}{V_k(\eta_0)} \frac{\partial V_k(\eta_0)}{\partial \eta} = -i\omega_k(\eta_0)$$

Using $\omega_k^2(\eta_0) \rightarrow k^2$ as $\eta_0 \rightarrow -\infty$ we immediately find

$$\delta\chi_k = \frac{1}{\sqrt{2k}} e^{-ik\eta}; \quad \eta \rightarrow -\infty$$

We call this the *Bunch-Davies vacuum*.²⁹ On super-horizon scales $-k\eta \ll 1$ we find

$$\delta\chi_k \simeq \frac{i}{\sqrt{2k}} \frac{aH}{k}$$

implying

$$|\delta\phi_k| \simeq \frac{H}{\sqrt{2k^3}}$$

²⁸Then for $\eta_0 \rightarrow -\infty$ all modes can be taken in the instantaneous minimal configuration.

²⁹This is simply the instantaneous vacuum in de Sitter for $\eta \rightarrow -\infty$.

9.3. The scalar field in quasi de Sitter

We consider the solutions valid for slow-roll inflation

$$\delta\chi_k = \sqrt{-\eta} [A_k H_\nu^{(1)}(-k\eta) + B_k H_\nu^{(2)}(-k\eta)]$$

We consider the sub-horizon limit $-k\eta \gg 1$. Note, in this limit

$$H_\nu^{(1)} \sim \sqrt{\frac{2}{-\pi k\eta}} \exp\left[i\left(-k\eta - \frac{\pi\nu_\phi}{2} - \frac{\pi}{4}\right)\right]; \quad H_\nu^{(2)} \sim \sqrt{\frac{2}{-\pi k\eta}} \exp\left[-i\left(-k\eta - \frac{\pi\nu_\phi}{2} - \frac{\pi}{4}\right)\right]$$

We normalize to the Bunch-Davies vacuum by requiring

$$\delta\chi_k = \frac{1}{\sqrt{2k}} e^{-ik\eta}; \quad -k\eta \gg 1$$

implying

$$A_k = \frac{\sqrt{\pi}}{2} \exp\left[i\left(\nu_\phi + \frac{1}{2}\right)\frac{\pi}{2}\right]; \quad B_k = 0$$

Then we consider the super-horizon limit $-k\eta \ll 1$. Note,

$$H_\nu^{(1)} \sim \sqrt{\frac{2}{\pi}} e^{-i\pi/2} 2^{\nu_\phi-3/2} \frac{\Gamma(\nu_\phi)}{\Gamma(3/2)} (-k\eta)^{-\nu_\phi}$$

implying

$$\delta\chi_k = \exp\left[i\left(\nu_\phi - \frac{1}{2}\right)\frac{\pi}{2}\right] 2^{\nu_\phi-3/2} \frac{\Gamma(\nu_\phi)}{\Gamma(3/2)} \frac{1}{\sqrt{2k}} (-k\eta)^{1/2-\nu_\phi}$$

and³⁰

$$|\delta\phi_k| \simeq \frac{H}{\sqrt{2k^3}} \left(\frac{k}{aH}\right)^{3/2-\nu_\phi}; \quad -k\eta \ll 1 \Rightarrow \frac{k}{aH} \ll 1$$

We see this reduces to the $m = 0$ case when $\nu_\phi = 3/2$.

9.4. The power spectrum

Consider a generic $g(\mathbf{x}, t)$ with

$$g(\mathbf{x}, t) = \int \frac{d^3k}{(2\pi)^{3/2}} e^{i\mathbf{k}\cdot\mathbf{x}} g_k(t)$$

We define the *power spectrum* through

³⁰We ignore some superfluous prefactors.

$$\langle 0 | g_{k_1}^* g_{k_2} | 0 \rangle \equiv \delta^{(3)}(\mathbf{k}_1 - \mathbf{k}_2) \frac{2\pi^2}{k^3} P_g(k_1)$$

Then

$$\begin{aligned} \langle 0 | |g^2(\mathbf{x}, t)| | 0 \rangle &= \int \frac{d^3 k_1 d^3 k_2}{(2\pi)^3} e^{i(\mathbf{k}_1 - \mathbf{k}_2) \cdot \mathbf{x}} \langle 0 | g_{k_1}^* g_{k_2} | 0 \rangle \\ &= \int \frac{d^3 k_1 d^3 k_2}{(2\pi)^3} e^{i(\mathbf{k}_1 - \mathbf{k}_2) \cdot \mathbf{x}} \delta^{(3)}(\mathbf{k}_1 - \mathbf{k}_2) \frac{2\pi^2}{k^3} P_g(k_1) \\ &= \int \frac{d^3 k_2}{(2\pi)^3} \frac{2\pi^2}{k_2^3} P_g(k_2) \\ &\stackrel{\text{spherical}}{=} \int \frac{dk}{k} P_g(k) \end{aligned}$$

Then the *variance* of $g(\mathbf{x}, t)$ is related to the power spectrum by

$$\langle 0 | |g^2(\mathbf{x}, t)| | 0 \rangle = \int \frac{dk}{k} P_g(k)$$

We can compute this for the scalar field

$$\begin{aligned} \langle 0 | |\delta\phi|^2 | 0 \rangle &= \left\langle 0 \left| \left| \int \frac{d^3 k}{(2\pi)^{3/2}} (\delta\phi_k a_k e^{i\mathbf{k} \cdot \mathbf{x}} + \delta\phi_k^* a_k^\dagger e^{-i\mathbf{k} \cdot \mathbf{x}}) \right|^2 \right| 0 \right\rangle \\ &= \left\langle 0 \left| \int \frac{d^3 k_1 d^3 k_2}{(2\pi)^3} |\delta\phi_k|^2 a_{k_1} a_{k_2}^\dagger \right| 0 \right\rangle \\ &= \int \frac{d^3 k_1 d^3 k_2}{(2\pi)^3} |\delta\phi_k|^2 \delta^{(3)}(\mathbf{k}_1 - \mathbf{k}_2) \\ &= \int \frac{d^3 k}{(2\pi)^3} |\delta\phi_k|^2 \\ &= \int \frac{dk}{k} \frac{k^3}{2\pi^2} |\delta\phi_k|^2 \\ &\stackrel{\text{by definition}}{=} \int \frac{dk}{k} P_{\delta\phi}(k) \end{aligned}$$

implying

$$P_{\delta\phi}(k) = \frac{k^3}{2\pi^2} |\delta\phi_k|^2$$

With the above we find

$$P_{\delta\phi}(k) = \left(\frac{H}{2\pi}\right)^2 \left(\frac{k}{aH}\right)^{3-2\nu_\phi}$$

The quantity $P_{\delta\phi}$ gives information on $V(\phi)$ and slow-roll inflation through the *spectral index*³¹

$$n_{\delta\phi} - 1 \equiv \frac{d \ln P_{\delta\phi}}{d \ln k} = 3 - 2\nu_\phi$$

Also,

$$\nu_\phi^2 = \frac{9}{4} + 3\epsilon - \frac{m^2}{H^2}$$

$$\stackrel{m^2=V_{\phi\phi}}{=} \frac{9}{4} + 3\epsilon - 3\eta_\phi$$

$$\nu_\phi = \frac{3}{2} \sqrt{1 + \frac{4}{3}\epsilon - \frac{4}{3}\eta_\phi}$$

$$\simeq \frac{3}{2} + \epsilon - \eta_\phi$$

implying

$$n_{\delta\phi} - 1 = 2\eta_\phi - 2\epsilon$$

which is quite nice! The analysis when we include fluctuations of the metric is more complicated, however, the effect only changes the value above slightly.³²

³¹Commonly referred to as *tilt*. This is one of the primary values we wish to determine in order to test different inflationary models.

³²We find $n_{\delta\phi} - 1 = 2\eta_\phi - 6\epsilon$. See below.

10. Metric Perturbations³³

Above we considered fluctuations of the scalar field $\delta\phi$. We ignored their effect on $T_{\mu\nu}$, however, they will induce fluctuations $\delta T_{\mu\nu}$. Assuming ϕ dominates³⁴ the EFE then imply we will have perturbations $\delta g_{\mu\nu}$ since

$$\delta\mathcal{R}_{\mu\nu} - \frac{1}{2}\delta(g_{\mu\nu}\mathcal{R}) = 8\pi G_N \delta T_{\mu\nu}$$

But, this also goes the other way, so

$$\delta\phi \Leftrightarrow \delta g_{\mu\nu}$$

10.1. The linear metric

We can expand $g_{\mu\nu}$ as

$$g_{\mu\nu} = g_{\mu\nu}^{(0)} + \delta g_{\mu\nu}(\mathbf{x}, t)$$

with $g_{\mu\nu}^{(0)}$ being the FRW metric. We write

$$g_{\mu\nu}(\mathbf{x}, \eta) = a^2(\eta)(\eta_{\mu\nu} + h_{\mu\nu})$$

with

$$g^{\mu\nu}(\mathbf{x}, \eta) = a^{-2}(\eta)(\eta^{\mu\nu} + h^{\mu\nu})$$

where $h^{\mu\nu} = \eta^{\mu\rho}\eta^{\nu\alpha}h_{\rho\alpha}$. We parametrize $h_{\mu\nu}$ as³⁵

$$h_{\mu\nu} \equiv \begin{pmatrix} -2\phi & B_i \\ B_i & -2\psi\delta_{ij} + E_{ij} \end{pmatrix}$$

with ϕ and ψ being scalars, B_i being a vector and E_{ij} being a traceless tensor.

We can decompose the vectors using the *Helmholtz decomposition*

$$u_i = \partial_i V + V_i$$

where $\partial_i V$ has no curl and V_i has no divergence. Then

$$B_i = B_i^{\text{scalar}} + B_i^{\text{vector}}$$

with

³³See <https://arxiv.org/pdf/hep-ph/0210162> for a review.

³⁴Which it would for inflation.

³⁵This parametrizes the metric degrees of freedom.

$$B_i^{\text{scalar}} = \partial_i B; \quad \partial^i B_i^{\text{vector}} = 0$$

Likewise

$$E_{ij} = E_{ij}^{\text{scalar}} + E_{ij}^{\text{vector}} + E_{ij}^{\text{tensor}}$$

with

$$E_{ij}^{\text{scalar}} = \left[\partial_i \partial_j - \frac{\delta_{ij}}{3} \nabla^2 \right] E$$

$$E_{ij}^{\text{vector}} = -\frac{1}{2} (\partial_j E_i + \partial_i E_j)$$

We want E_{ij} to be traceless. This implies E_i has no divergence $\partial^i E_i = 0$. Also,

$$\partial_i E_{ij}^{\text{tensor}} = 0; \quad E^{\text{tensor}^i}_i = 0$$

We can now determine the total degrees of freedom

$$\text{dof} = \text{scalars} + \underbrace{\text{vector}}_{B_i^{\text{vector}} \text{ and } E_i} \times 2 + \underbrace{\text{tensor}}_{E_{ij}^{\text{tensor}}} = 10$$

Note, at linear order these evolve independently.

10.2. Gauge transformations

Consider the change of coordinates due to an infinitesimal gauge transformation

$$\tilde{x}^\mu = x^\mu + \delta x^\mu$$

We decompose $\delta x^\mu = \xi^\mu(x^\nu)$ as

$$\delta x^0 = \xi^0(x^\mu); \quad \delta x^i = \xi^i(x^\mu) + \partial^i \xi(x^\mu); \quad \partial_i \xi^i = 0$$

Then we find two scalar gauge degrees of freedom ξ^0 and ξ . Which we can use to eliminate two scalar degrees of freedom above.

Consider a scalar of the form

$$q(\eta, \mathbf{x}) = q_0(\eta) + \delta q(\eta, \mathbf{x})$$

and apply the gauge transformation $\eta \rightarrow \tilde{\eta} = \eta + \xi^0(\eta, \mathbf{x})$ ³⁶

$$\delta q(\tilde{\eta}, \mathbf{x}) = \delta q(\eta, \mathbf{x}) + \frac{\partial q_0(\eta)}{\partial \eta} \xi^0$$

³⁶ Comes from $\tilde{q}(x) = q(x - \xi) = q(x) - \xi^\mu \partial_\mu q$

Then $ds^2 \stackrel{!}{=} d\tilde{s}^2$ implies

$$a^2(\tilde{x}^0)(1 + 2\tilde{\phi})(d\tilde{x}^0)^2 \stackrel{!}{=} a^2(x^0)(1 + 2\phi)(dx^0)^2$$

with

$$a^2(\tilde{x}^0)^2 = a^2(x^0) + 2a\partial_\eta a \xi^0$$

$$d\tilde{x}^0 \stackrel{\text{by chain-rule}}{=} (1 + \partial_\eta \xi^0) dx^0 + \partial_i \xi^0 dx^i$$

implying

$$\tilde{\phi} = \phi - \frac{\partial \xi^0}{\partial \eta} - \frac{1}{a} \frac{\partial a}{\partial \eta} \xi^0$$

Likewise,

$$\tilde{B} = B + \xi^0 + \frac{\partial \xi}{\partial \eta}$$

$$\tilde{\psi} = \psi - \frac{1}{3} \nabla^2 \xi + \frac{1}{a} \frac{\partial a}{\partial \eta} \xi^0$$

$$\tilde{E} = E + 2\xi$$

10.3. Gauge invariants

We would like gauge invariant quantities since these are nice to work with. An example are the *Bardeen potentials*

$$\Phi = -\phi + \frac{1}{a} \frac{\partial}{\partial \eta} \left[\left(-B + \frac{1}{2} \frac{\partial E}{\partial \eta} \right) a \right]$$

$$\Psi = -\psi - \frac{1}{6} \nabla^2 E + \frac{1}{a} \frac{\partial a}{\partial \eta} \left(B - \frac{1}{2} \frac{\partial E}{\partial \eta} \right)$$

Also, consider³⁷

$${}^{(3)}\mathcal{R} = \frac{4}{a^2} \nabla^2 \psi$$

which is the *spatial curvature* on hypersurfaces of $\eta = \text{constant}$. However, the *curvature*

³⁷Note, we assume $k = 0$.

perturbation ψ does depend on the gauge. We instead consider³⁸

$$\mathcal{R} = \psi + \frac{1}{a} \frac{\partial a}{\partial \eta} \left(\frac{\partial \phi}{\partial \eta} \right)^{-1} \delta \phi = \psi + H \frac{\delta \phi}{\dot{\phi}}$$

which is gauge invariant. We can pick the *comoving gauge* $\delta \phi = 0$ where

$$\mathcal{R} = \psi|_{\delta \phi=0}$$

Then $\mathcal{R}$ is actually the curvature perturbation on a comoving slice! Likewise we can define

$$\zeta = \psi + \frac{1}{a} \frac{\partial a}{\partial \eta} \left(\frac{\partial \rho}{\partial \eta} \right)^{-1} \delta \rho = \psi + H \frac{\delta \rho}{\dot{\rho}}$$

and pick the *uniform density gauge* $\delta \rho = 0$ where

$$\zeta = \psi|_{\delta \rho=0}$$

Then ζ is the curvature perturbation on spatial slices of uniform ρ .

We would like to understand the construction of these quantities. We do this by finding the gauge transformation bringing us to the comoving gauge by picking ξ^0 as to cancel $\delta \phi$. We have

$$\psi \rightarrow \psi + \frac{1}{a} \frac{\partial a}{\partial \eta} \delta \tau$$

under $t \rightarrow t + \delta \tau$. Where $\delta \tau \equiv \xi^0$ and $\xi = 0$. Also,

$$\delta \phi \rightarrow \delta \phi - \frac{\partial \phi}{\partial \eta} \delta \tau = 0 \Rightarrow \delta \tau = \left(\frac{\partial \phi}{\partial \eta} \right)^{-1} \delta \phi$$

$$\delta \rho \rightarrow \delta \rho - \frac{\partial \rho}{\partial \eta} \delta \tau = 0 \Rightarrow \delta \tau = \left(\frac{\partial \rho}{\partial \eta} \right)^{-1} \delta \rho$$

This implies ψ transforms as

$$\psi \rightarrow \psi + \frac{1}{a} \frac{\partial a}{\partial \eta} \left(\frac{\partial \phi}{\partial \eta} \right)^{-1} \delta \phi$$

$$\psi \rightarrow \psi + \frac{1}{a} \frac{\partial a}{\partial \eta} \left(\frac{\partial \rho}{\partial \eta} \right)^{-1} \delta \rho$$

³⁸We use ϕ to mean the scalar field.

as we had above.

We can also pick the *flat gauge* $\psi_{\text{flat}} = 0$. Then

$$\delta\tau = -\frac{\psi}{\mathcal{H}}$$

where $\mathcal{H}$ is a *Hubble-like* parameter

$$\mathcal{H} \equiv \frac{1}{a} \frac{\partial a}{\partial \eta}$$

implying

$$\delta\phi \rightarrow \delta\phi + \frac{1}{\mathcal{H}} \frac{\partial \phi}{\partial \eta} \psi \equiv Q$$

with Q being the *Mukhanov-Sasaki variable*

$$Q = \delta\phi + \frac{1}{\mathcal{H}} \frac{\partial \phi}{\partial \eta} \psi$$

$$= \delta\phi + \frac{\dot{\phi}}{H} \psi$$

$$\equiv \frac{\dot{\phi}}{H} \mathcal{R}$$

which is gauge invariant and

$$Q = \delta\phi|_{\psi=0}$$

10.4. The longitudinal gauge

We choose $B = E = 0$. This is called the *longitudinal gauge*. The Bardeen potentials become

$$\Phi = -\phi; \quad \Psi = -\psi$$

so ϕ and ψ become gauge invariant. Also, assuming $T_{ij} = 0$ for $i \neq j$ the EFE imply

$$\psi = \phi$$

We also find from T_{0i} ³⁹

$$\dot{\psi} + H\psi = 4\pi G_N \dot{\phi} \delta\phi$$

³⁹We now use ϕ to mean the scalar field.

$$= \epsilon H^2 \frac{\delta\phi}{\dot{\phi}}$$

and from T_{00}

$$-3H(\dot{\psi} + H\psi) + \frac{\nabla^2\psi}{a^2} = 4\pi G_N [\dot{\phi}\delta\dot{\phi} - \dot{\phi}^2\psi - V_\phi\delta\phi]$$

and from T_{ii}

$$-\left[2\frac{\ddot{a}}{a} + \frac{\dot{a}^2}{a^2}\right]\psi - 3H\dot{\psi} - \ddot{\psi} = -[\dot{\phi}\delta\dot{\phi} - \dot{\phi}^2\psi - V_\phi\delta\phi]$$

Then using

$$\dot{H} = 4\pi G_N \dot{\phi}^2; \quad \ddot{\phi} + 3H\dot{\phi} + V_\phi = 0$$

we can obtain the trace of the EFE

$$\ddot{\psi}_k + \left[H + 2\frac{\ddot{\phi}}{\dot{\phi}}\right]\dot{\psi}_k + 2\left[\dot{H} - H\frac{\ddot{\phi}}{\dot{\phi}}\right]\psi_k + \frac{k^2}{a^2}\psi_k = 0$$

or

$$\frac{\partial^2\psi_k}{\partial\eta^2} + 2\mathcal{H}(\eta_\phi - \epsilon)\frac{\partial\psi_k}{\partial\eta} + 2\mathcal{H}^2(\eta_\phi - 2\epsilon)\psi_k + k^2\psi_k = 0$$

Assuming super-horizon scales and $\epsilon, \eta_\phi \rightarrow 0$ we have

$$\frac{\partial^2\psi_k}{\partial\eta^2} = 0 \Rightarrow a^2[H\dot{\psi}_k + \ddot{\psi}_k] = 0$$

implying $\dot{\psi}_k \simeq \ddot{\psi}_k \simeq 0$. Then

$$\psi_k = \epsilon H \frac{\delta\phi_k}{\dot{\phi}}$$

and

$$\mathcal{R}_k \simeq H \frac{\delta\phi_k}{\dot{\phi}}$$

Then

$$P_{\mathcal{R}}(k) = \frac{k^3}{2\pi^2} \frac{H^2}{\dot{\phi}^2} |\delta\phi_k|^2 = \frac{k^3}{4\pi^2} \frac{1}{M_p} \frac{1}{\epsilon} |\delta\phi_k|^2$$

Also, the Klein-Gordon equation becomes

$$\delta\ddot{\phi}_k + 3H \delta\dot{\phi}_k + \frac{k^2}{a^2} d\phi_k + V_{\phi\phi} \delta\phi_k = \underbrace{-2\psi_k V_\phi + 4\dot{\psi}_k \dot{\phi}_c}_{\text{terms from } \delta g}$$

with

$$|4\dot{\psi}_k \dot{\phi}_c| \ll |\psi_k V_\phi|; \quad \psi_k = \epsilon H \frac{\delta\phi_k}{\dot{\phi}_c}; \quad V_\phi \simeq -3H \dot{\phi}_c$$

implying

$$\delta\ddot{\phi}_k + 3H \delta\dot{\phi}_k + (V_{\phi\phi} + 6\epsilon H^2) \delta\phi_k = 0$$

Then with $\chi_k \equiv \delta\phi_k/a$ we find

$$\frac{\partial^2 \delta\chi_k}{\partial \eta^2} - \frac{1}{\eta^2} \left(\nu^2 - \frac{1}{4} \right) \delta\chi_k = 0$$

with

$$\nu^2 = \frac{9}{4} + 9\epsilon - 3\eta_\phi$$

Then on super-horizon scales

$$|\delta\phi_k| \simeq \frac{H}{\sqrt{2k^3}} \left(\frac{k}{aH} \right)^{\frac{3}{2}-\nu}$$

as before. This implies

$$P_{\mathcal{R}}(k) = \frac{1}{2M_p^2 \epsilon} \left(\frac{H}{2\pi} \right)^2 \left(\frac{k}{aH} \right)^{n_{\mathcal{R}}-1}$$

where

$$n_{\mathcal{R}} - 1 = \frac{d \ln P_{\mathcal{R}}}{d \ln k} = 3 - 2\nu = 2\eta_\phi - 6\epsilon$$

Also, we can see

$$\dot{\mathcal{R}}_k \simeq \dot{\zeta}_k \rightarrow 0$$

implying the comoving perturbation is conserved on super-horizon scales. Note, we can compute

$$P_\Phi \stackrel{\text{radiation dominated}}{=} \left(\frac{2}{3} \right)^2 P_{\mathcal{R}}$$

$$\text{matter dominated} \stackrel{=}{=} \left(\frac{3}{5}\right)^2 P_{\mathcal{R}}$$

and

$$\frac{\delta T}{T} = \frac{1}{3}\Phi = \frac{1}{5}\mathcal{R}$$

11. The Unruh Effect

We have seen how the *natural* vacuum state can change with time. We will now see how the *natural* vacuum can also be completely observer dependent, even in Minkowski spacetime.

11.1. Rindler spacetime

We consider two observers in Minkowski spacetime. One is at rest while the other is accelerated by some constant acceleration. We would like to relate the vacuum states of these two observers.

We need a *comoving* coordinate system for the accelerated observer. The metric describing the rest frame of the accelerated observer is called Rindler spacetime. Assuming the acceleration is along x we have

$$ds^2 \stackrel{\text{Minkowski}}{=} -dt^2 + dx^2$$

We transform our coordinates as⁴⁰

$$t = \frac{1}{a} e^{a\xi} \sinh a\tau$$

$$x = \frac{1}{a} e^{a\xi} \cosh a\tau$$

implying

$$ds^2 = e^{2a\xi} (d\tau^2 - d\xi^2)$$

Consider an observer along $x^\mu(\tau)$. Then

$$u^\mu u_\mu = -1; \quad u^\mu = \frac{dx^\mu}{d\tau}$$

and

$$a^\mu a_\mu \equiv \alpha^2; \quad a^\mu = \frac{du^\mu}{d\tau}$$

Within the observers rest frame we have $u^\mu = (1, 0)$ implying $(u^0)^2 = 1$ so

$$\frac{d}{d\tau}(u^\mu u_\mu) = 0$$

⁴⁰Here (τ, ξ) are the rest frame coordinates.

implying $a^0 = 0$ so α is the proper acceleration as seen by the observer and $a^\mu(\tau) = (0, \alpha)$. Then

$$u^0 = \sqrt{1 + (u^1)^2}; \quad \frac{du^1}{d\tau} = \alpha \sqrt{1 + (u^1)^2}$$

implying

$$\frac{dx}{d\tau} = \sinh \alpha \tau$$

$$\frac{dt}{d\tau} = \cosh \alpha \tau$$

or

$$x(\tau) = \frac{1}{\alpha} \cosh \alpha \tau$$

$$t(\tau) = \frac{1}{\alpha} \sinh \alpha \tau$$

Which we recognise above. Then the metric for a uniformly accelerated observer in Minkowski spacetime with

$$\frac{1}{\alpha} \equiv \frac{1}{a} e^{a\xi}$$

is

$$ds^2 = e^{2a\xi} (-d\tau^2 + d\xi^2)$$

Assuming $\xi = \text{constant}$ we have

$$x^2 - t^2 = \frac{1}{\alpha^2}$$

which is a hyperbole. Also, there are now horizons at $x = \pm t$. We call these the Rindler horizons.

11.2. Quantum fields in Rindler space

Consider the action for a scalar field with $m = 0$

$$S[\phi] = -\frac{1}{2} \int d^2x \sqrt{-g} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi$$

which is conformally invariant under

$$g_{\mu\nu} \rightarrow \tilde{g}_{\mu\nu} = \Omega^2 g_{\mu\nu}$$

We have

$$S[\phi] \stackrel{\text{rest Minkowski}}{=} \frac{1}{2} \int dt dx [(\partial_t \phi)^2 - (\partial_x \phi)^2]$$

$$S[\phi] \stackrel{\text{comoving Rindler}}{=} \frac{1}{2} \int d\tau d\xi [(\partial_\tau \phi)^2 - (\partial_\xi \phi)^2]$$

so

$$\frac{\partial^2 \phi}{\partial t^2} - \frac{\partial^2 \phi}{\partial x^2} = 0; \quad \tilde{U}_k = \frac{1}{\sqrt{2k}} e^{ikx - i\omega t}$$

$$\frac{\partial^2 \phi}{\partial \tau^2} - \frac{\partial^2 \phi}{\partial \xi^2} = 0; \quad U_k = \frac{1}{\sqrt{2k}} e^{ik\xi - i\omega \tau}$$

We obtain two mode expansions

$$\phi(t, x) \stackrel{\text{rest Minkowski}}{=} \int \frac{dk}{\sqrt{2\pi}} \frac{1}{\sqrt{2|k|}} [a_k e^{-i|k|t + ikx} + a_k^\dagger e^{i|k|t - ikx}]$$

$$\phi(\tau, \xi) \stackrel{\text{comoving Rindler}}{=} \int \frac{dk}{\sqrt{2\pi}} \frac{1}{\sqrt{2|k|}} [b_k e^{-i|k|\tau + ik\xi} + b_k^\dagger e^{i|k|\tau - ik\xi}]$$

implying

$$a_k |0_M\rangle = 0; \quad b_k |0_R\rangle = 0$$

with $|0_M\rangle \neq |0_R\rangle$.

We now define *light-cone coordinates*

$$\bar{u} = t - x; \quad \bar{v} = t + x$$

$$u = \tau - \xi; \quad v = \tau + \xi$$

Then

$$ds^2 = -d\bar{u}d\bar{v} = -e^{a(v-u)} du dv$$

We also have

$$\frac{\partial^2 \phi(\bar{u}, \bar{v})}{\partial \bar{u} \partial \bar{v}} = 0; \quad \frac{\partial^2 \phi(u, v)}{\partial u \partial v} = 0$$

implying

$$\phi(\bar{u}, \bar{v}) = A(\bar{u}) + B(\bar{v})$$

$$\phi(u, v) = P(u) + Q(v)$$

We would like these functions. Using $\omega = |k|$ for the Minkowski fields we have

$$\phi(\bar{u}, \bar{v}) = \int_0^\infty \frac{d\omega}{\sqrt{2\pi}} \frac{1}{\sqrt{2\omega}} \left[\underbrace{a_\omega e^{-i\omega\bar{u}} + a_\omega^\dagger e^{i\omega\bar{u}}}_{A(\bar{u})} + \overbrace{a_{-\omega} e^{-i\omega\bar{v}} + a_{-\omega}^\dagger e^{i\omega\bar{v}}}^{B(\bar{v})} \right]$$

and taking $\Omega = |k|$ for the Rindler fields we have

$$\phi(u, v) = \int_0^\infty \frac{d\Omega}{\sqrt{2\pi}} \frac{1}{\sqrt{2\Omega}} \left[\underbrace{a_\Omega e^{-i\Omega u} + a_\Omega^\dagger e^{i\Omega u}}_{P(u)} + \overbrace{a_{-\Omega} e^{-i\Omega v} + a_{-\Omega}^\dagger e^{i\Omega v}}^{Q(v)} \right]$$

Consider

$$\phi(u, v) = A(\bar{u}(u)) + B(\bar{v}(v))$$

implying

$$A(\bar{u}(u)) = P(u)$$

or

$$\int \frac{d\omega}{\sqrt{2\pi}} \frac{1}{\sqrt{2\omega}} [a_\omega e^{-i\omega\bar{u}} + a_\omega^\dagger e^{i\omega\bar{u}}] = \int \frac{d\Omega}{\sqrt{2\pi}} \frac{1}{\sqrt{2\Omega}} [b_\Omega e^{-i\Omega u} + b_\Omega^\dagger e^{i\Omega u}]$$

We Fourier transform in u . The RHS becomes

$$\int \frac{du}{\sqrt{2\pi}} e^{i\Omega u} P(u) = \frac{1}{\sqrt{2|\Omega|}} \begin{cases} b_\Omega, & \Omega > 0 \\ b_{|\Omega|}^\dagger, & \Omega < 0 \end{cases}$$

and the LHS becomes

$$\int \frac{du}{\sqrt{2\pi}} e^{i\Omega u} A(\bar{u}) = \int_0^\infty \frac{d\omega}{\sqrt{2\omega}} [a_\omega F(\omega, \Omega) + a_\omega^\dagger F(-\omega, \Omega)]$$

with

$$F(\omega, \Omega) = \int \frac{du}{2\pi} e^{i\Omega u - i\omega\bar{u}} = \int \frac{du}{2\pi} \exp \left[i\Omega u + \frac{i\omega}{a} e^{-au} \right]$$

We find by inspection

$$b_\Omega = \int_0^\infty d\omega [\alpha_{\omega\Omega} a_\omega + \beta_{\omega\Omega} a_\omega^\dagger]$$

with

$$\alpha_{\omega\Omega} = \sqrt{\frac{\Omega}{\omega}} F(\omega, \Omega); \quad \beta_{\omega\Omega} = \sqrt{\frac{\Omega}{\omega}} F(-\omega, \Omega)$$

Also, we have

$$[a_\omega, a_{\omega'}^\dagger] = \delta(\omega - \omega'); \quad [b_\Omega, b_{\Omega'}^\dagger] = \delta(\Omega - \Omega')$$

which implies

$$\int d\omega [\alpha_{\omega\Omega} \alpha_{\omega\Omega'}^* - \beta_{\omega\Omega} \beta_{\omega\Omega'}^*] = \delta(\Omega - \Omega')$$

Then

$$\begin{aligned} \langle N_\Omega \rangle &= \langle 0_M | b_\Omega^\dagger b_\Omega | 0_M \rangle \\ &= \int d\omega d\omega' \left\langle 0_M \left| \left[\underbrace{\alpha_{\omega\Omega}^* a_\omega^\dagger + \beta_{\omega\Omega}^* a_\omega}_0 \right] \left[\underbrace{\alpha_{\omega'\Omega} a_{\omega'} + \beta_{\omega'\Omega} a_{\omega'}^\dagger}_0 \right] \right| 0_M \right\rangle \\ &= \int d\omega d\omega' \langle 0_M | \beta_{\omega\Omega}^* \beta_{\omega'\Omega} a_\omega a_{\omega'}^\dagger | 0_M \rangle \\ &= \int d\omega |\beta_{\omega\Omega}|^2 \\ &= n_\Omega \delta(0) \end{aligned}$$

11.3. The Unruh temperature

We had

$$\beta_{\omega\Omega} = \sqrt{\frac{\Omega}{\omega}} F(-\omega, \Omega)$$

Consider

$$F(\omega, \Omega) = \int \frac{du}{2\pi} \exp \left[i\Omega u + \frac{i\omega}{a} e^{-au} \right]$$

We substitute

$$t = -\frac{i\omega}{a}e^{-au}; \quad dt = -at \, du$$

Then

$$\begin{aligned} F(\omega, \Omega) &= -\frac{1}{2\pi a} \int \frac{dt}{t} \exp \left[\frac{i\Omega}{a} \ln \left(\frac{\omega}{iat} \right) - t \right] \\ &= -\frac{1}{2\pi a} \int \frac{dt}{t} \exp \left[\frac{i\Omega}{a} \left(\ln \frac{\omega}{a} - \frac{i\pi}{2} - \ln t \right) - t \right] \\ &= -\frac{1}{2\pi a} \exp \left[\frac{i\Omega}{a} \ln \frac{\omega}{a} + \frac{\pi\Omega}{2a} \right] \int dt t^{-1-i\Omega/a} e^{-t} \\ &= \underbrace{\quad}_{\text{contour-dependent}} \frac{1}{2\pi a} \exp \left[\frac{i\Omega}{a} \ln \frac{\omega}{a} + \frac{\pi\Omega}{2a} \right] \Gamma \left(-\frac{i\Omega}{a} \right) \end{aligned}$$

We have

$$\ln(-\omega) = \ln(-1) + \underbrace{\ln \omega}_{i\pi}$$

implying

$$F(\omega, \Omega) = F(-\omega, \Omega) e^{\pi\Omega/a}$$

Then

$$\begin{aligned} \delta(\Omega - \Omega') &= \int_0^\infty \frac{d\omega}{\omega} \sqrt{\Omega\Omega'} [F(\omega, \Omega) F^*(\omega, \Omega') - F(-\omega, \Omega) F^*(-\omega, \Omega')] \\ &= \left\{ \exp \left[\frac{\pi\Omega + \pi\Omega'}{a} \right] - 1 \right\} \int_0^\infty \frac{d\omega}{\omega} \sqrt{\Omega\Omega'} F(-\omega, \Omega) F^*(-\omega, \Omega') \end{aligned}$$

implying

$$\int_0^\infty \frac{d\omega}{\omega} \sqrt{\Omega\Omega'} F(-\omega, \Omega) F^*(-\omega, \Omega') = \left\{ \exp \left[\frac{2\pi\Omega}{a} \right] - 1 \right\}^{-1} \delta(\Omega - \Omega')$$

Then

$$\langle N_\Omega \rangle = \int_0^\infty d\omega |\beta_{\omega\Omega}|^2$$

$$\begin{aligned}
&= \int_0^\infty \frac{d\omega}{\omega} \Omega |F(-\omega, \Omega)|^2 \\
&= \left\{ \exp \left[\frac{2\pi\Omega}{a} \right] - 1 \right\}^{-1} \delta(0)
\end{aligned}$$

so

$$n_\Omega = \left\{ \exp \left[\frac{2\pi\Omega}{a} \right] - 1 \right\}^{-1}$$

which is the Bose-Einstein distribution!

We compare this distribution with the thermal spectrum of a $m = 0$ particle with energy $E = \Omega$

$$n(E) = \left\{ \exp \left(\frac{E}{T} \right) - 1 \right\}^{-1}$$

where T is the *Unruh temperature*

$$T_{\text{Unruh}} = \frac{a}{2\pi}$$

Then a uniformly accelerated observer in Minkowski spacetime experiences a thermal spectrum with the temperature T_{Unruh} . This is called the *Unruh effect*.

With all units we have

$$k_B T_{\text{Unruh}} = \frac{a\hbar}{2\pi c} \sim \text{insanely small}$$

implying the Unruh effect is very inefficient. However, it can become strong in certain domains.

12. Hawking Radiation

We know staying outside a black hole requires an acceleration. Then any observer which is not in free-fall experiences the emission of radiation. This is referred to as *Hawking radiation*.

12.1. Rindler-esque derivation of Hawking radiation

Consider the Schwarzschild metric

$$ds^2 = \left(1 - \frac{2MG}{r}\right) dt^2 - \left(1 - \frac{2MG}{r}\right)^{-1} dr^2 - r^2 d\Omega^2$$

We see the metric diverges at the Schwarzschild radius r_* defined by

$$r_* \equiv 2MG$$

However, the curvature is finite at r_* so this is not a *true* singularity. We refer to the surface $r = 0$ as the true singularity. Any observer falling into the black hole would experience nothing special at r_* . But, to a distant observer the infalling observer would *freeze* at the horizon, eventually disappearing due to redshift. Also, as $r \rightarrow r_*$ we find $c_{\text{apparent}} \rightarrow 0$. This implies nothing can escape upon crossing the horizon. This is why we call them *black*.

We define ρ by

$$\rho = \int_{r_*}^r \sqrt{g_{rr}(r')} dr'$$

Then

$$ds^2 = \left(1 - \frac{2MG}{r(\rho)}\right) dt^2 - d\rho^2 - r(\rho)^2 d\Omega^2$$

Close to the horizon we have

$$\rho \simeq \sqrt{r(r - 2MG)} + 2MG \sqrt{\frac{r}{2MG} - 1}$$

implying

$$ds^2 \simeq \rho^2 \left(\frac{dt}{4MG} \right)^2 - d\rho^2 - r^2(\rho) d\Omega^2$$

We define ξ by

$$\rho = \frac{1}{a} e^{a\xi}$$

Then

$$ds^2 = e^{2a\xi} (dt^2 - d\xi^2)$$

with

$$a = \frac{1}{4MG}$$

We recognise this as the Rindler metric for an accelerated observer with proper acceleration

$$\alpha = a e^{-a\xi}$$

With $\alpha = a$ we have $\xi = 0$.

We then immediately find the *Hawking temperature* by

$$\begin{aligned} T_H &= \frac{a}{2\pi} \\ &= \frac{1}{8\pi MG} \end{aligned}$$

implying black holes evaporate over time.

Note, we refer to the vacuum for the free-falling observer as the *Kruskal vacuum* $|0_K\rangle$ and the vacuum for an observer at constant r as the *Boulware vacuum* $|0_B\rangle$. These are analogous to the Minkowski vacuum $|0_M\rangle$ and the Rindler vacuum $|0_R\rangle$.

12.2. in de Sitter spacetime

We can write the de Sitter metric as

$$ds^2 = \left(1 - \frac{r^2}{r_{dS}^2}\right) dt^2 - \left(1 - \frac{r^2}{r_{dS}^2}\right)^{-1} dr^2 - r^2 d\Omega^2$$

where

$$r_{dS} \equiv H^{-1}$$

is the de Sitter radius. We can as before determine

$$T_{dS} = \frac{H}{2\pi} \sim \delta\phi_k$$

12.3. Thermodynamics of black holes

We consider black holes to be spherical bodies with radius r_* and surface temperature T_H . We can then write the luminosity of a black hole using the Stefan-Boltzmann law

$$L = \gamma \sigma T_H^4 A$$

Then

$$\frac{dM}{dt} = -L$$

implying

$$M(t) = M_0 \left(1 - \frac{t}{t_L}\right)^{1/3}$$

with

$$t_L = \frac{5120\pi M_0^3}{\gamma} \sim \text{lifetime}$$

where M_0 is the initial black hole mass. We see $t_L \sim M_0^3$ so black holes have to be quite small for this effect to be noticable.

We can determine the entropy of a black hole using

$$dE = dM = T_H dS_{\text{BH}} \sim 1^{\text{st}} \text{ law}$$

Assuming $S_{\text{BH}} \propto A$ as Bekenstein did we write

$$S_{\text{BH}} = \alpha 16\pi M^2$$

and find

$$\alpha \stackrel{!}{=} \frac{1}{4}$$

Which gives the Bekenstein-Hawking entropy

$$S_{\text{BH}} = \frac{A}{4} \sim \frac{k_B A}{4\ell_p^2}$$

which is weird since we normally consider $S = \ln N$ with N being the number of indistinguishable microstates.

We also find the black hole equation of state

$$E = \frac{1}{8\pi T_H}$$

implying

$$C_{\text{BH}} = \frac{\partial E}{\partial T} = -\frac{1}{8\pi T^2} < 0$$

So black holes become colder upon absorbing heat!

The *area theorem* states

$$\frac{dA}{dt} \geq 0$$

in any *classical* interaction. Where we note Hawking radiation is a quantum process. This is related to the *generalised 2nd law of thermodynamics*

$$dS_{\text{total}} = dS_{\text{matter}} + dS_{\text{BH}} \geq 0$$

which is just the 2nd law of thermodynamics where we include S_{BH} .

12.4. Consequences

Consider collapsing ordinary matter to form a black hole. Before collapse we have $S \propto V$ while after collapse we have $S_{\text{BH}} \propto A$. This implies a loss of information and a violation of the GLS given V is large enough. This is referred to as the *information paradox* and is usually resolved by either sacrificing unitarity or locality.

This also leads to the notion of *holography* which sacrifices locality and postulates

$$S_{\text{matter}} < \frac{A}{4}$$

i.e. the entropy of matter is bounded by the area which encloses it. Then as the matter is collapsed to form a black hole we have no issues.